SAFETY PRECAUTIONS

X	Stopped engine
X	Block the starting mechanism
X	Shut off starting air supply
X	Engage turning gear
X	Shut off cooling water
X	Shut off fuel oil
X	Shut off lubricating oil
	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D02-02	Piston rod/crosshead, tightening torque	430	Nm
D02-07	Piston rod/crosshead, tightening torque+angle	100/30	Nm/°
D02-20	Piston complete	921	kg
D04-01	Crosshead bearing, top clearance max.	0.55	mm
D04-02	Crosshead bearing, top clearance min.	0.15	mm
D04-03	Oil wedge length, L	10	mm
D04-04	Crosshead bearing cap	160	kg
D04-05	Crosshead bearing shell, upper	13	kg
D04-06	Crosshead bearing shell, lower	28	kg
D04-07	Crosshead bearing cap with bearing shell	173	kg
D04-21	Crosshead complete	1100	kg
D04-51	Connecting rod complete	1400	kg
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400 -1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P**909**51 refers to chapter 909.

Plate	Item No.	Description
P90451	47	Wire guide
P90451	59	Lifting attachment - connecting rod
P90451	72	Chain for suspending piston
P90451	96	Bracket for support, crosshead
P90451	106	Bracket for support, crosshead
P90461		Connecting Rod - Hydraulic Tools
P91351	10	Hydraulic pump, pneumatically operated
P91351	46	Hose with unions (1500 mm), complete
P91351	58	Hose with unions (3000 mm), complete
P91351	117	5-way distributor block, complete
P91356		Lifting Tools, Etc.
P91359		Torque Spanners
P91366	50	Feeler gauge

Tin-Aluminium bearings

The top clearance between the journal and a new bearing shell is the result of a summation of the production tolerances of the bearing assembly components.

The top clearance for a new bearing will normally be in the range stated in Data. Note that the figures are to be used for guidance only.

For the top clearance of a specific bearing, see the measurement in the Adjustment Sheet in Volume I, OPERATION.

1. Open the crankcase door at the relevant cylinder.

2. Turn the crankthrow concerned to BDC.

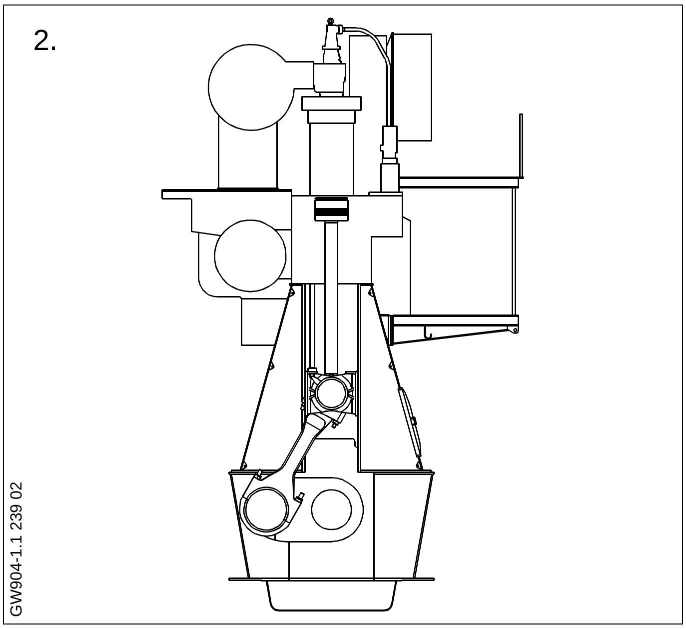
3. Measure the clearance in the crosshead bearing by inserting a feeler gauge between the bearing cap and the crosshead journal at the top of the upper bearing shell, at both sides and fore and aft. *See Data.*

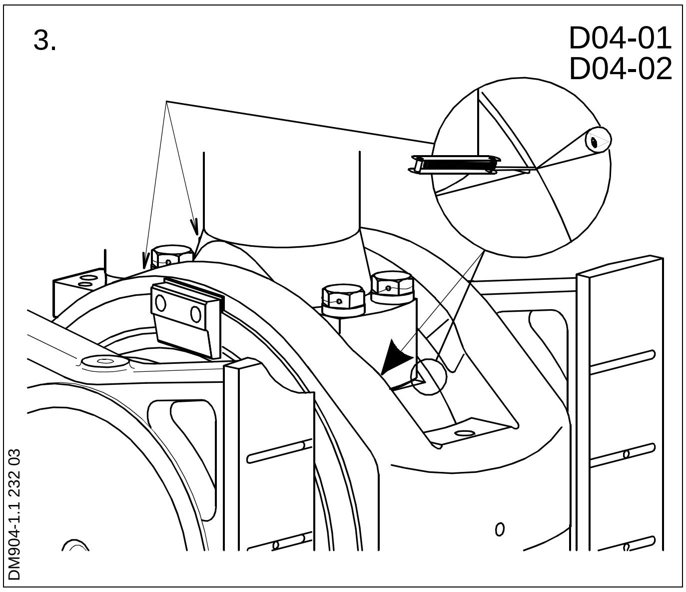
4. The difference between the **actual** clearance measurement and the measurement recorded in the Adjustment Sheet (or the clearance noted for a new bearing installed later) **must not** exceed 0.1 mm. If so, the crosshead bearing must be disassembled for inspection.
   See Procedure 904-1.2.

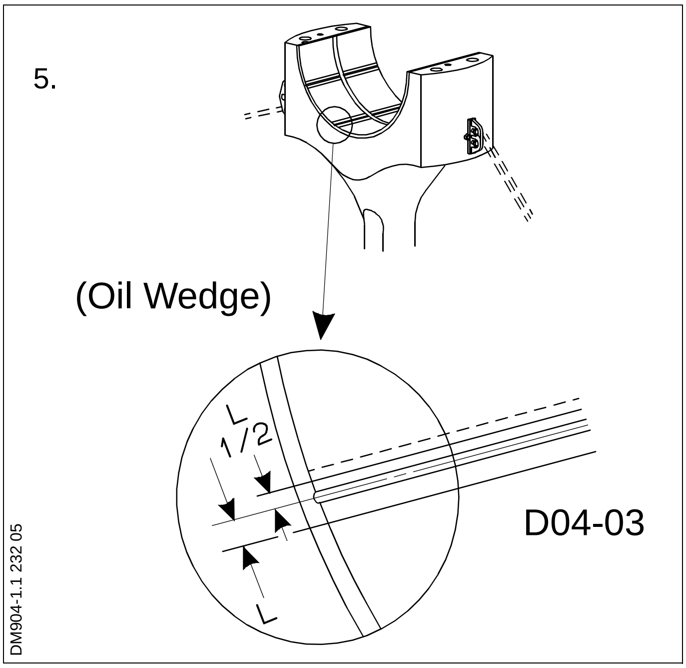
   For evaluation of the bearing shell, see Chapter 708, 'Bearings' in the instruction book, Volume I, OPERATION.

5. The wear limit for a crosshead bearing shell is confined to a 50% reduction of the oil wedge length **L**.

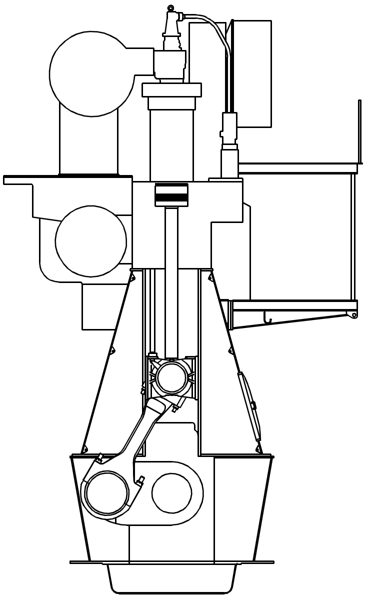
   If the wear limit exceeds the 50% reduction, the bearing shell must be replaced by a new one.





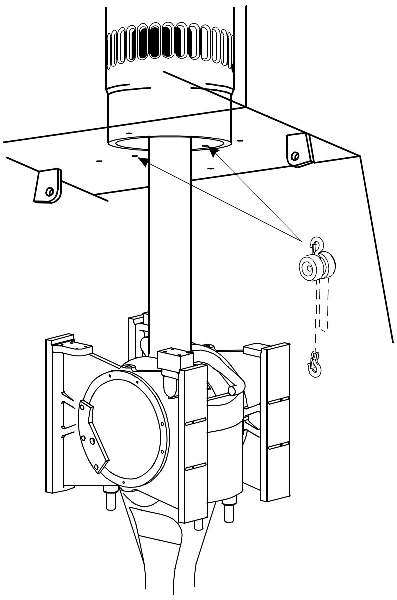


1.



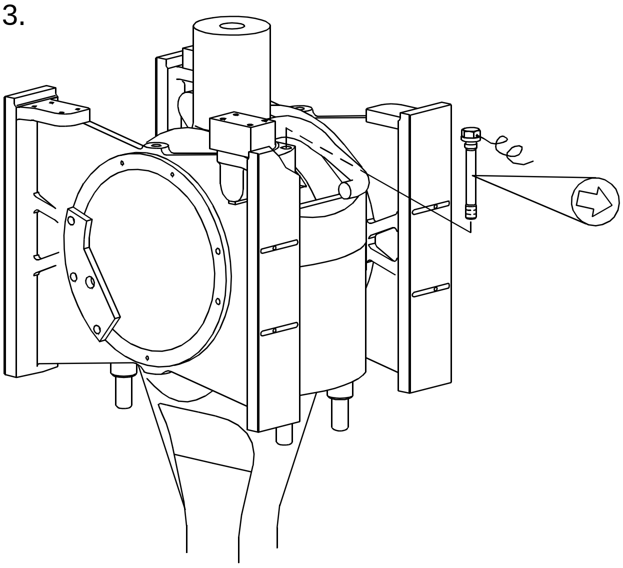
GW 904-1.1 239 02

2.



HN904-1.2 222 02

3.



HN904-1.2 222 03

This procedure applies to the following two dismantling situations:

- with piston mounted
- with piston removed

With piston mounted:

1. Turn the crankshaft down far enough to give access to the nuts and screws on the piston rod.

2. Mount two chains in the inner screw holes in the top of the crankcase, in the athwarthship direction, for suspending the piston rod.

3. Loosen and remove the locking wire from the screws on the piston rod foot. Remove the screws.

4. Mount a lifting eye bolt on each side of the piston rod.

5. Turn the crosshead to TDC.

   Hook the chains to the lifting eye bolts in the piston rod.

   Turn the crosshead downward, and the piston rod will then remain suspended from the two chains.

With piston removed:

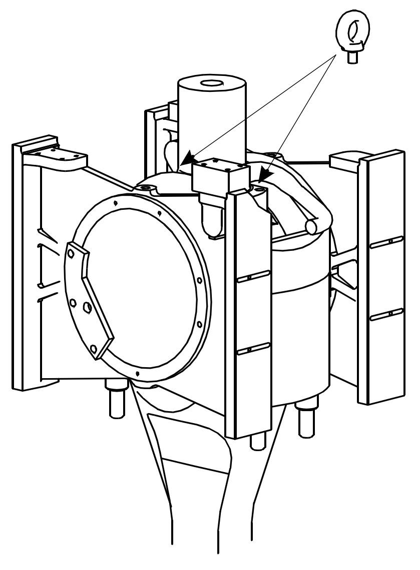
6. Turn to BDC.

7. Place the spacer rings around the nuts and screw the hydraulic jacks on to the studs.

   Loosen the crosshead bearing cap nuts. *For operation of the hydraulic jacks, see Procedure 913.*

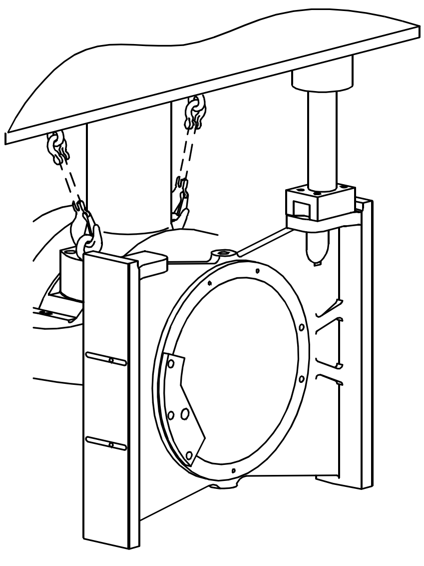
   Remove the hydraulic jacks and the spacer rings, and unscrew the nuts.

4.



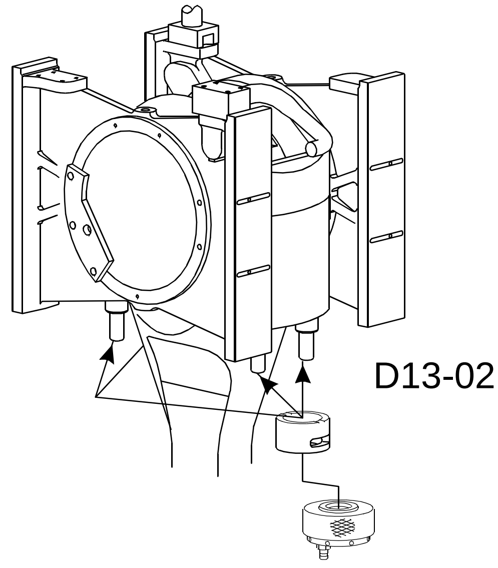
HN904-1.2 222 04

5.

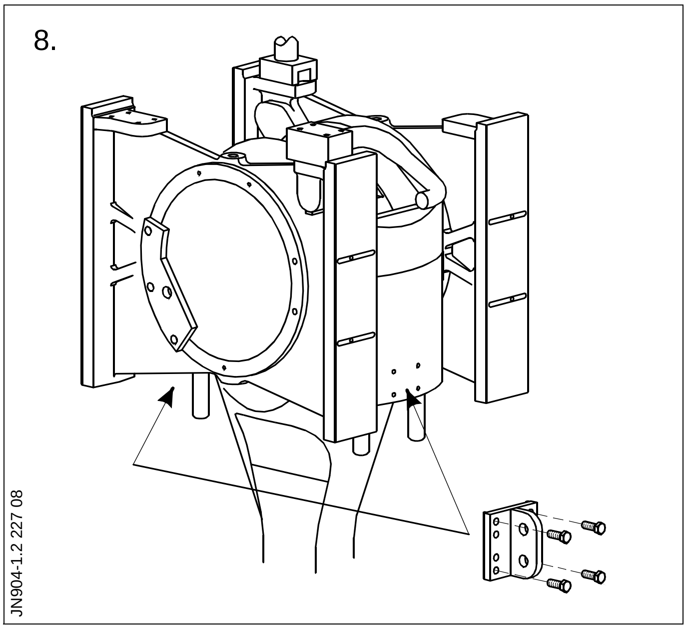


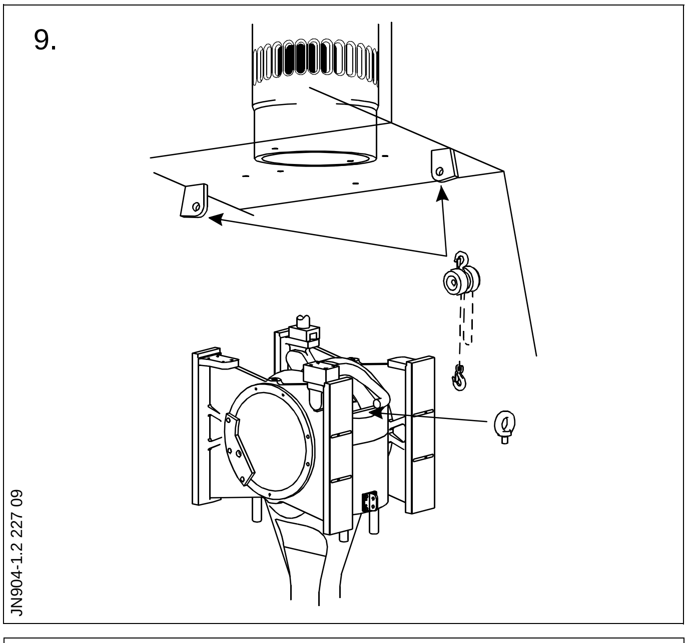
HN904-1.2 222 05

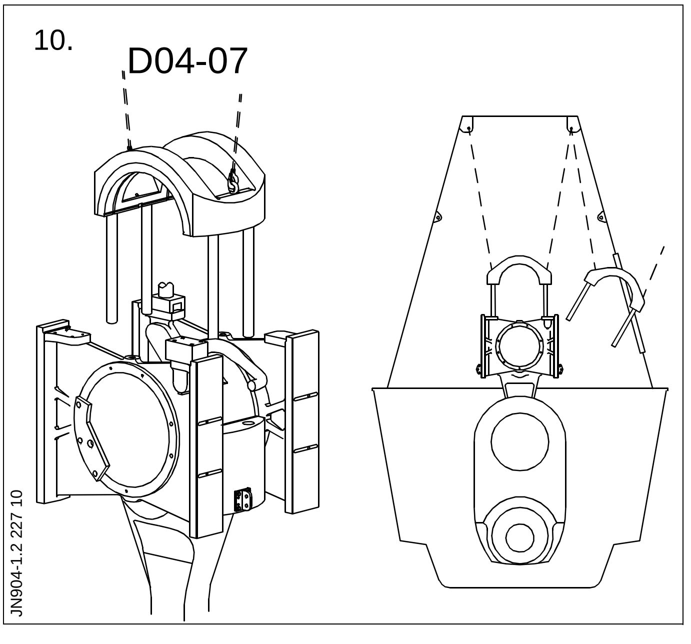
7.



HN904-5.2 217 01







8. Mount the lifting attachments on the head of the connecting rod.

9. Suspend two tackles from the lifting brackets, in the athwarthship direction.

10. Mount the wire guide at the top of the crankcase door frame to prevent damage.

    Mount two eye bolts in the top of the crosshead bearing cap.

    Hook the tackles on to the eye bolts, and remove the bearing cap from the engine. Check the upper part of the journal.

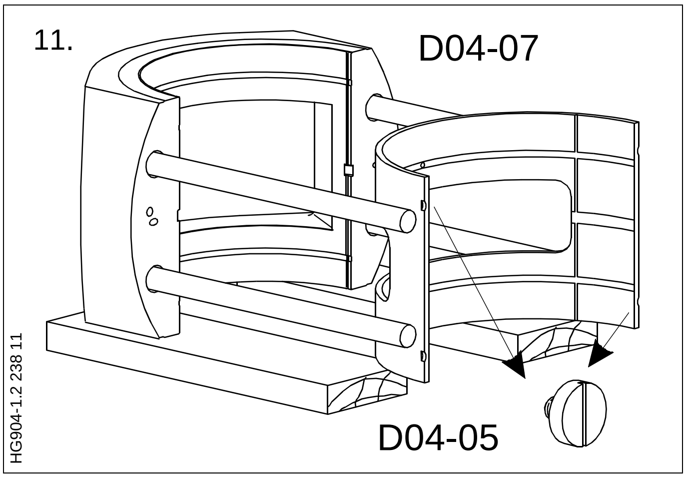
11. Place the bearing cap on one side on a couple of wooden planks.

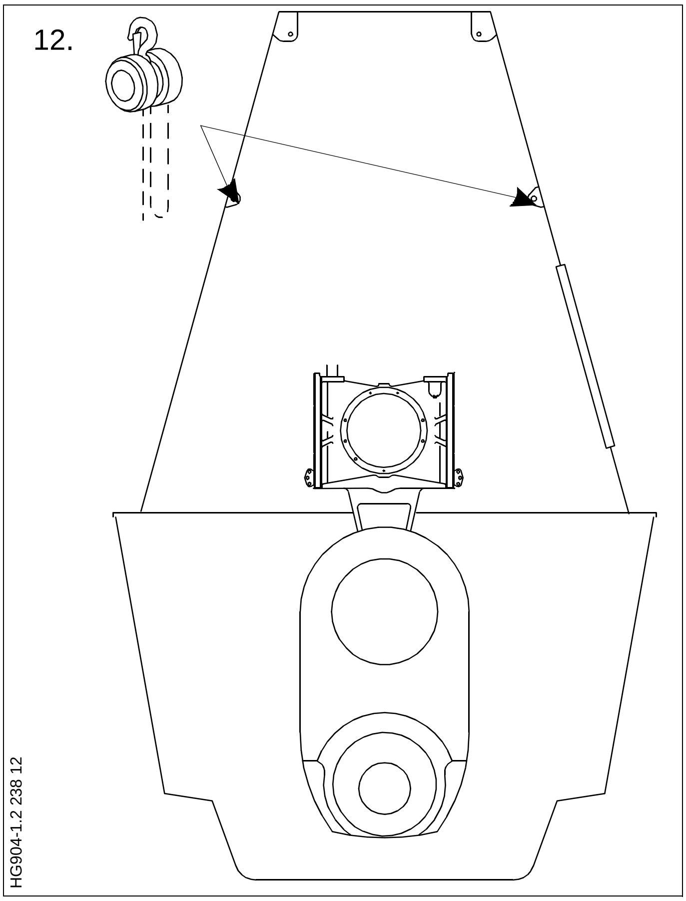
    Check the bearing shell, *see Procedure 904-1.1.*

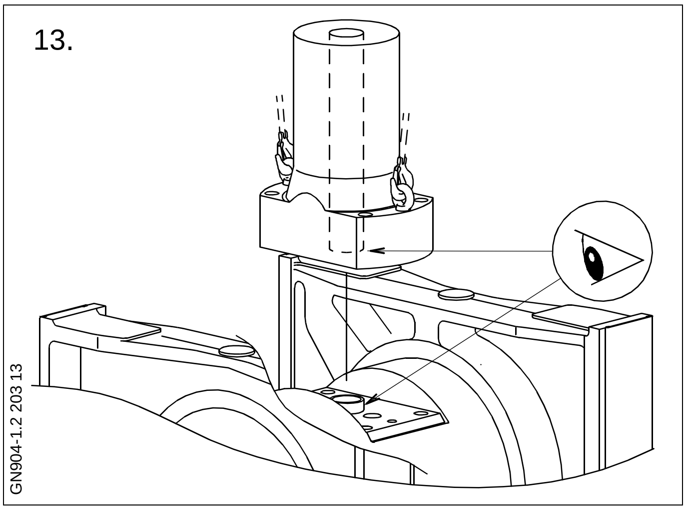
12. Fasten tackles to the fixed lifting brackets on the frame box wall.

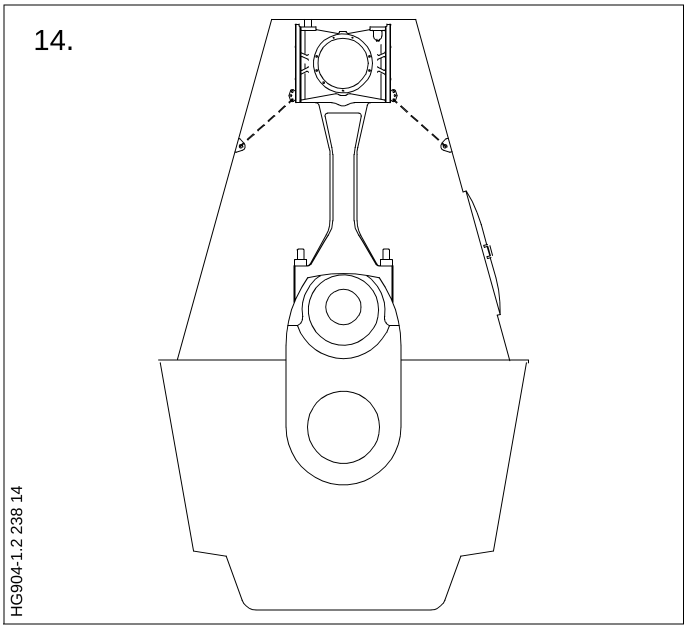
13. Turn the crosshead upwards until the piston rod lands on the crosshead. Ensure that the guide ring in the crosshead fits correctly in the centre hole of the piston rod.

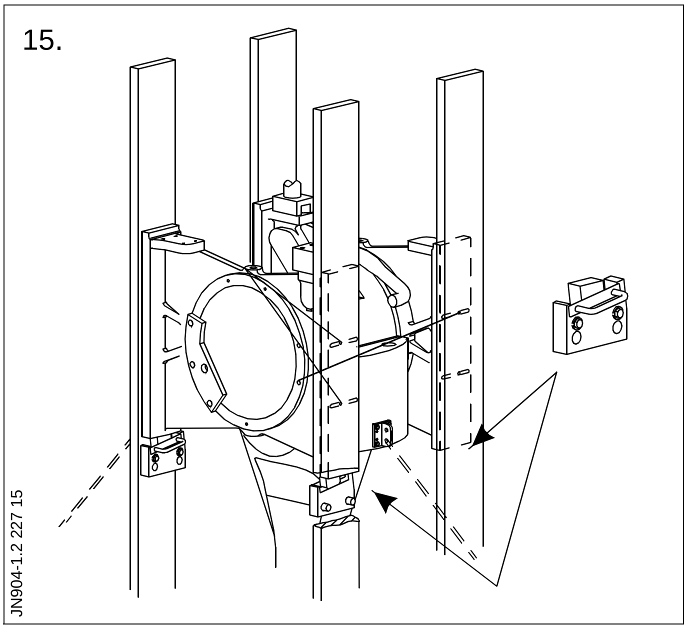
    Do not remove the chains or lifting eye bolts.

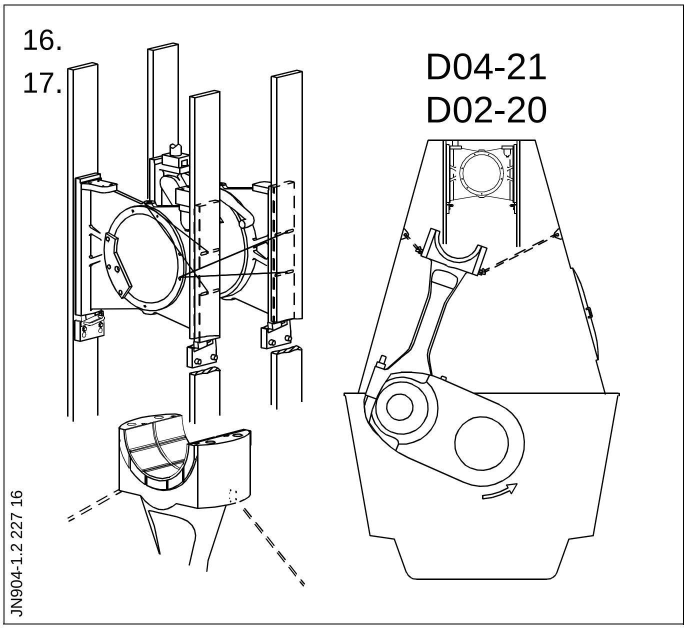












14. Turn to TDC and attach the tackle hooks to the lifting attachments. Haul the tackles tight.

15. Mount the four supports for guide shoes on the crosshead guides.

    Carefully turn the crank down towards the exhaust side, until the guide shoes rest on the supports.

    Adjust the support brackets to the guide shoes so that the weight of the crosshead is evenly distributed on the four supports.

    Haul the tackles tight.

16. Turn the crankthrow carefully towards BDC while ‘following’ with the tackles, thus continuously supporting the connecting rod.

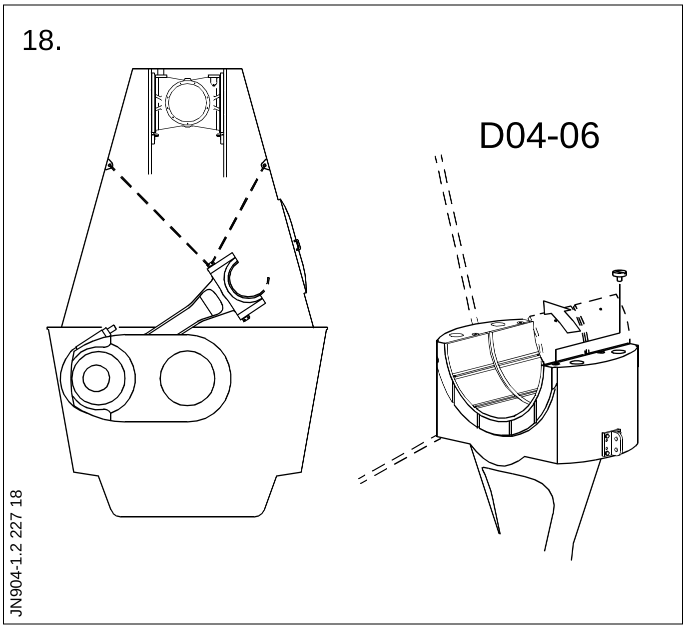
17. With the crosshead resting on the supports, check the lower part of the crosshead journal and the lower bearing shell.

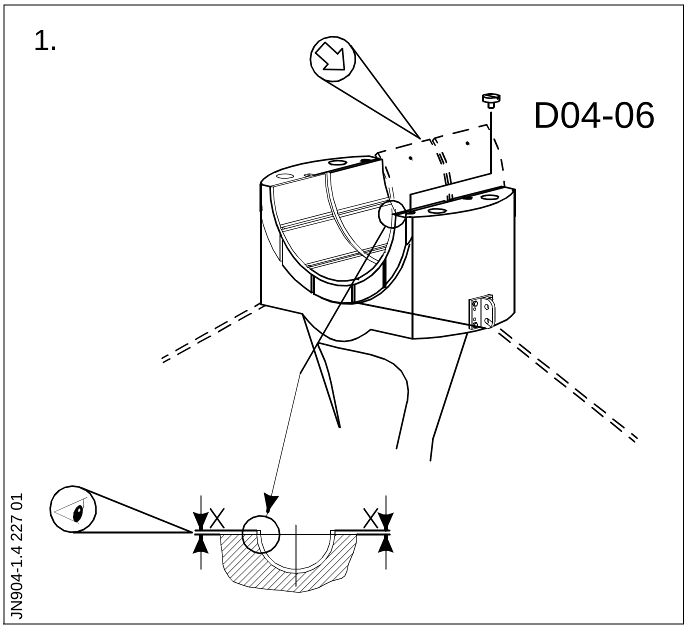
    Regarding checking of journal and bearing shells, see Volume I, OPERATION, Chapter 708, ‘Bearings’.

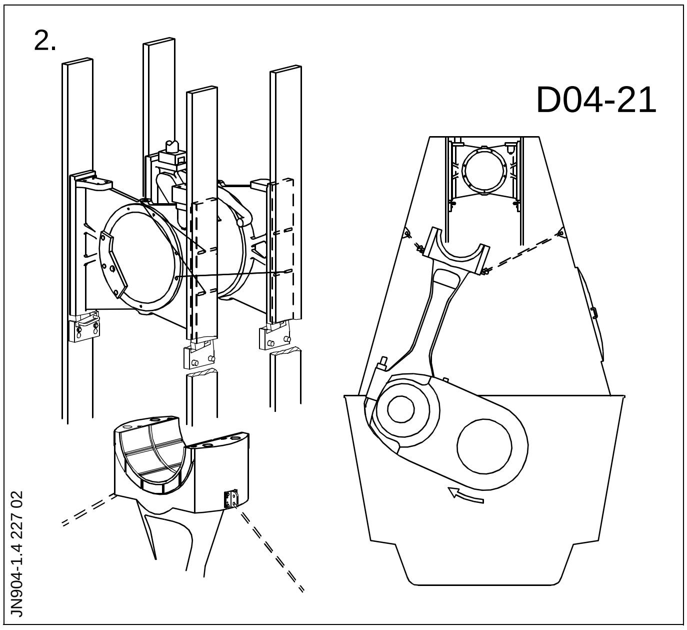
18. In cases where it is necessary to remove the lower bearing shell, tilt the connecting rod towards the doorway on the fuel pump side, using the tackles.

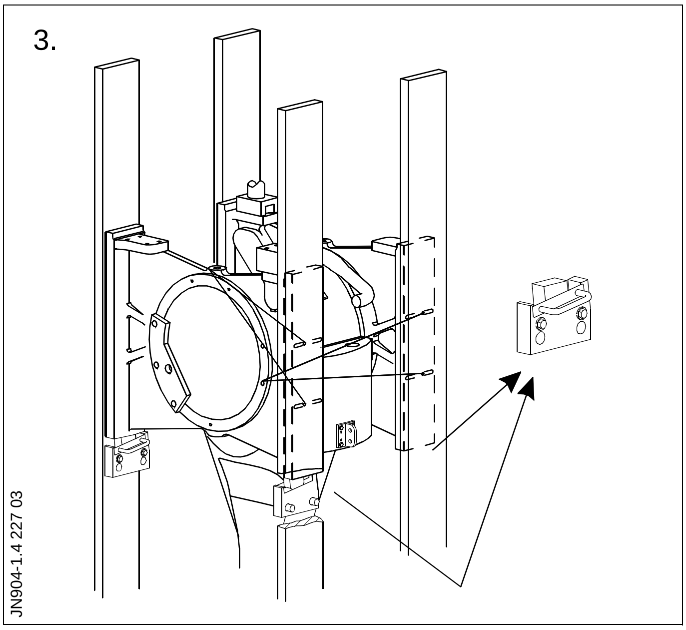
    Dismount the locking screws, and turn the bearing shell so far up that an eye bolt can be mounted.

    Lift the bearing shell out of the engine.









With piston mounted/
With piston removed:

1. Mount and secure the bearing shell in the bearing housing.

   The excess height **X** is to ensure the correct tightening-down of the bearing shell and **must not** be eliminated.

2. Raise the connecting rod to an upright position.

   Turn to TDC while ‘following’ with the tackles, for assembling the crosshead and the connecting rod.

   Take care that the guide shoes do not damage the bearing shell.

3. Remove the supports from the crosshead guides.

   Remove the tackles from the crosshead.

   Turn the crank throw to BDC.

   If the piston is mounted, slowly turn down until it is fully suspended from the chains.

4. Lift the bearing cap into the engine. Lower the bearing cap onto the crosshead and remove the tackles. Remove the lifting attachments from the connecting rod and the wire guide from the door frame.

Note!
Take care that the bearing studs do not damage the crosshead.

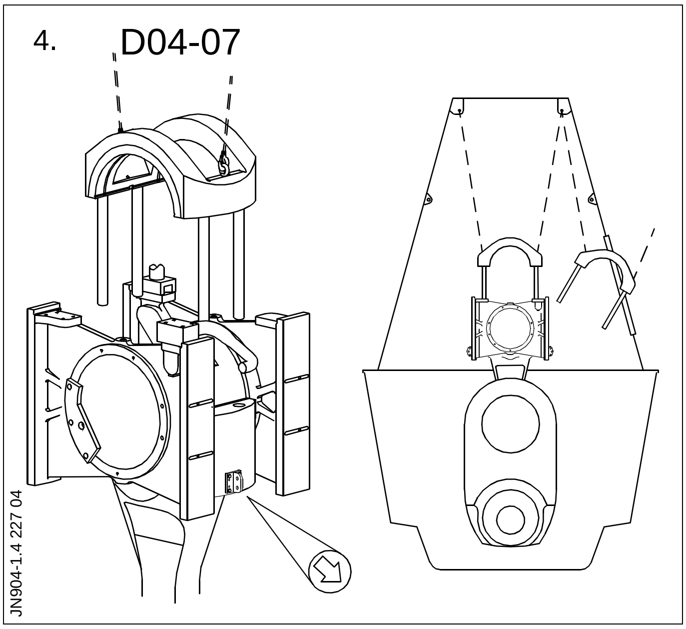
5. Tighten all four crosshead bearing cap nuts simultaneously. *See Data.*
   For operation of hydraulic jacks, see Section 913.

6. Mount the piston.
   See Procedure 902-1.4.

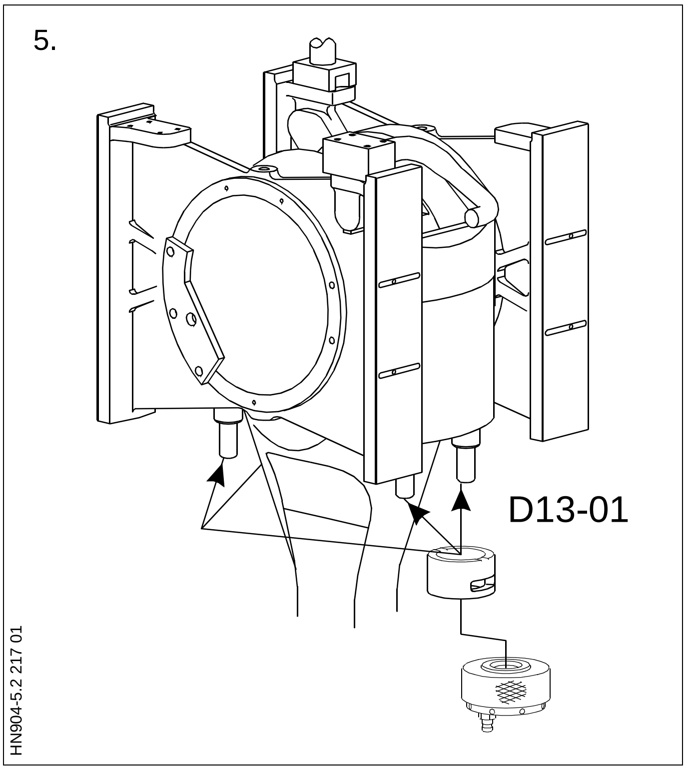
With piston mounted:

7. Turn the crosshead upwards until the piston rod lands on the crosshead. Ensure that the guide ring in the crosshead fits correctly in the centre hole of the piston rod.

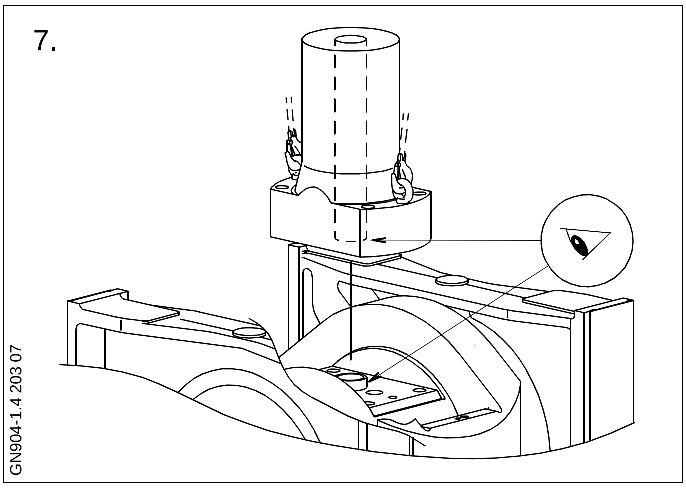
4.



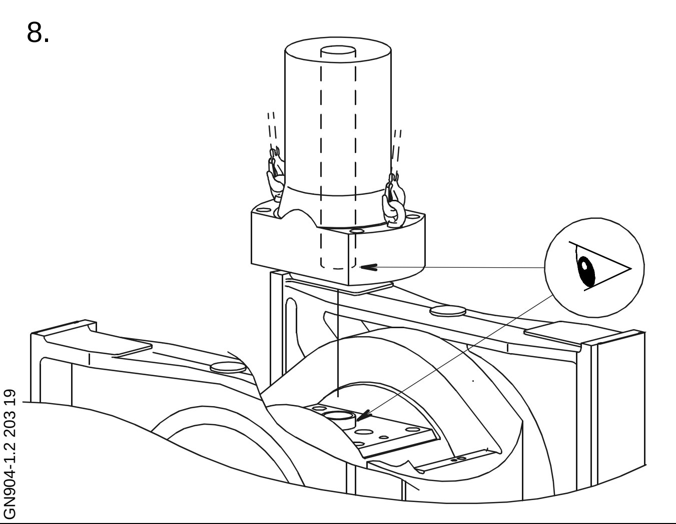
5.



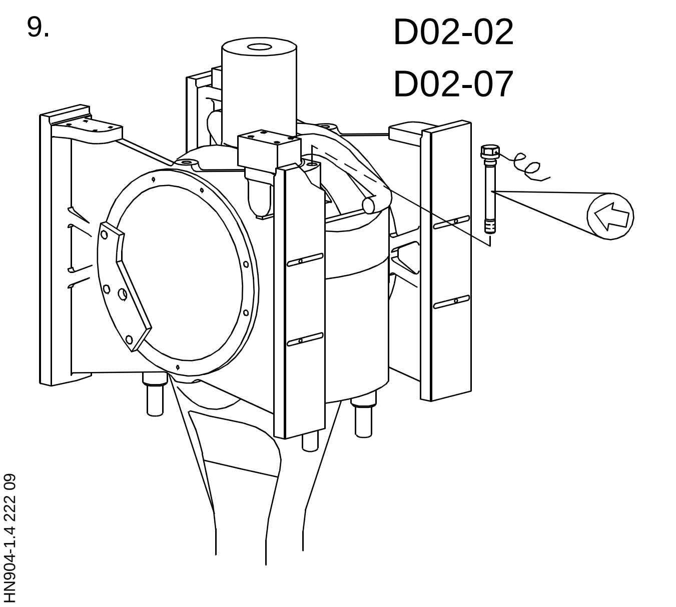
7.



8.



9.



8. Unhook the chains from the lifting eye bolts in the piston rod. Remove the chains and eye bolts from the top of the crankcase, from the piston rod and from the crosshead bearing cap.

9. Turn down to BDC, tighten the screws in the piston rod, and lock with locking wire. *See Data.*

   Mount the locking wire in such a way that the wire is tightened if a screw works loose. *See Procedure 913-7.*

SAFETY PRECAUTIONS

X	Stopped engine
X	Block the starting mechanism
X	Shut off starting air supply
X	Engage turning gear
X	Shut off cooling water
	Shut off fuel oil
X	Shut off lubricating oil
	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D04-04	Crosshead bearing cap	160	kg
D04-08	Cooling oil outlet pipe, tightening torque	50	Nm
D04-12	Guide strip screws, tightening torque	100	Nm
D04-13	Guide plate, tightening torque	135	Nm
D04-15	Telescopic pipe, tightening torque	50	Nm
D04-19	Telescopic pipe, tightening angle	25	°
D04-21	Crosshead complete	1100	kg
D04-22	Guide shoe	159	kg
D04-23	Cooling oil outlet pipe	9	kg
D04-50	Connecting rod, without bearing caps	1020	kg
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400 -1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P**909**51 refers to chapter 909.

Plate	Item No.	Description
P90451	47	Wire guide
P90451	59	Lifting attachment - connecting rod
P90451	60	Lifting tools - crosshead
P90451	72	Chain for suspending piston
P90451	84	Retaining tool telescopic pipe
P90451	214	Torque wrench offset tool
P90462	11	Guide shoe extractor, complete
P90463		Crosshead - Hydraulic Tools
P91351	10	Hydraulic pump, pneumatically operated
P91351	46	Hose with unions (1500 mm), complete
P91351	58	Hose with unions (3000 mm), complete
P91351	117	5-way distributor block, complete
P91356		Lifting Tools, Etc.
P91359		Torque Spanners
P91366	50	Feeler gauge

1. Dismount the piston.
   See Procedure 902-1.2.

2. Dismount the main bearing lubricating oil pipes.

3. Dismount the cooling oil outlet pipe from the guide shoe and the drain oil slotted pipe.

   Loosen and remove the screws which secure the telescopic pipe to the guide shoe.

   In order to reach the screw in the corner behind the telescopic pipe, use the offset tool along with a socket wrench.

4. Mount the retaining tool for the telescopic pipe on the stuffing box housing for the telescopic pipe.

   Turn the crosshead to TDC.

   Suspend the telescopic pipe by means of the tool.

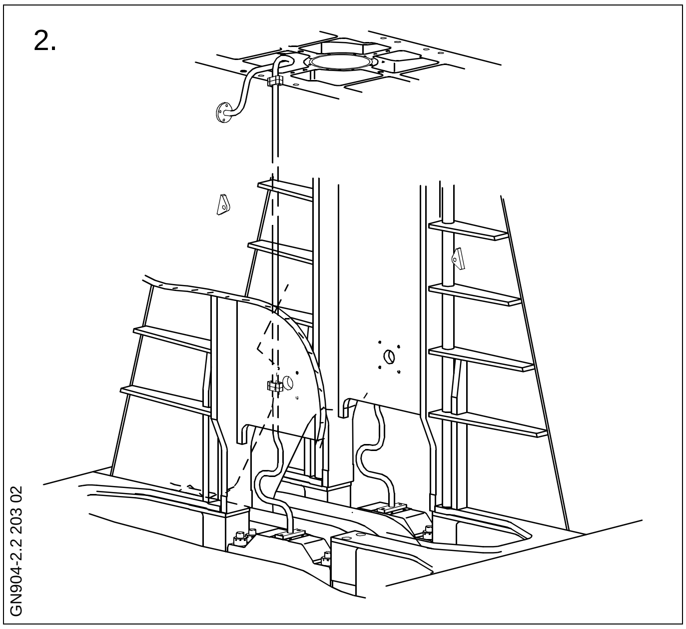
   Turn the crosshead to gain access to the nuts on the crosshead bearing studs.

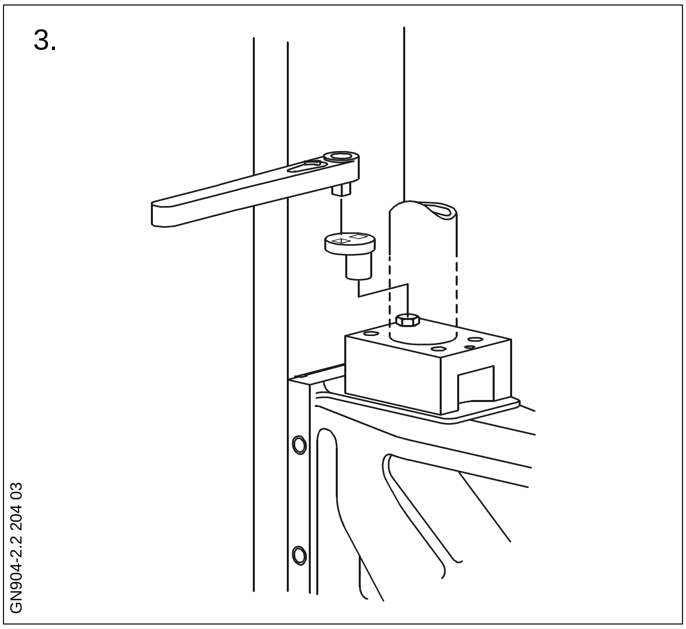
   Mount two eye bolts in the top of the crankcase in the fore-and-aft direction.

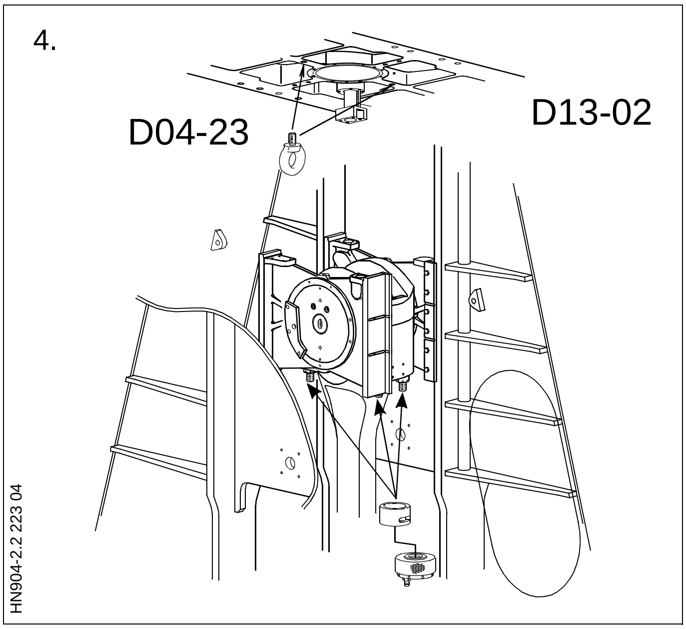
   Mount the spacer rings and the hydraulic jacks for loosening the nuts on the crosshead bearing studs.

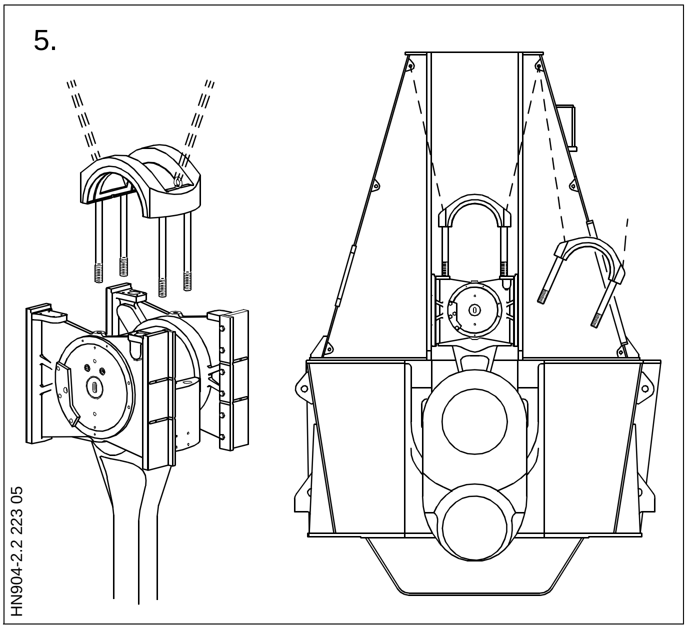
   For operation of the hydraulic tools, see Section 913.

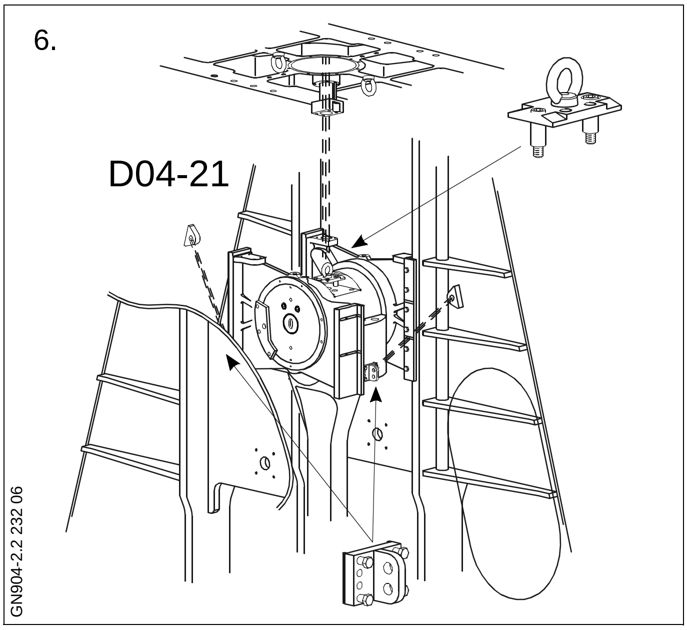
   Loosen the nuts, remove the hydraulic jacks and unscrew the nuts.

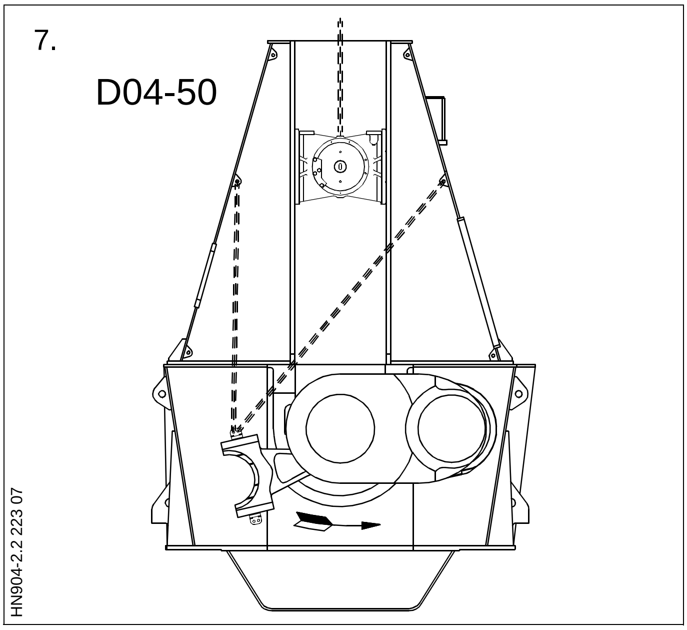












5. Mount eye bolts in the crosshead bearing cap and hook the tackles on to the lifting brackets in the top of the crankcase.

   Dismount the bearing cap and lift it out of the engine.

6. Mount the special lifting tool on the cross-head.

   Mount the lifting attachments for fixing the connecting rod, on the head of the connect-ing rod.

   Fasten tackles to the lifting brackets on the frame box wall and attach the tackle hooks to the lifting attachments. Haul the tackles tight.

   Attach the flat-plaited wire strap to the en-gine room crane.

   Hook the engine room crane on to the lifting tool on the crosshead, and lift the cross-head.

7. Using the tackles, tilt the connecting rod to-wards the exhaust side, while turning the crankthrow towards the camshaft side.

   Transfer the tackles from one lifting attach-ment to another as necessary.

   When the crankthrow is 90° after BDC, stop turning.

   By alternate use of the tackles, tilt the con-necting rod until it rests against a couple of wooden planks in the bottom of the bed-plate.

8. Lower the crosshead to a position just above the main bearing caps.

   Remove the guide strips and both guide plates from the guide shoes.

   Mount lifting eye bolts in both guide shoes.

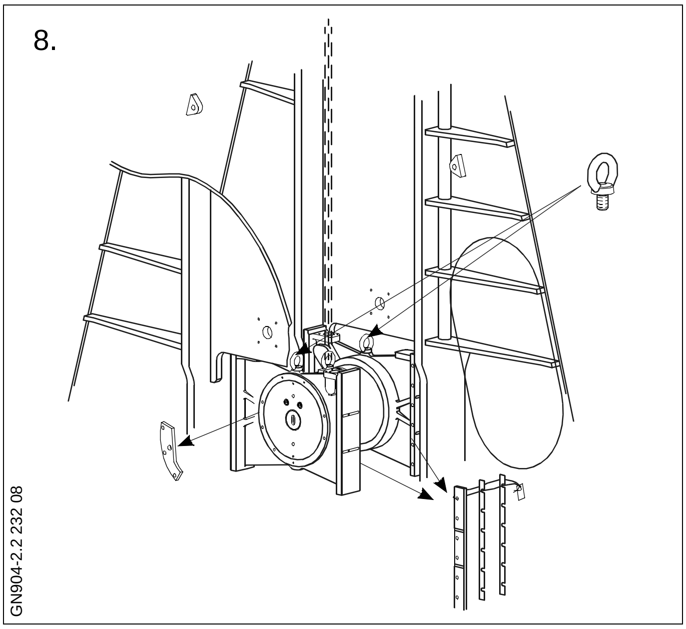
9. Suspend two tackles from the eye bolts in the top of the crankcase, in the fore-and-aft direction, and attach the tackles to the guide shoes.

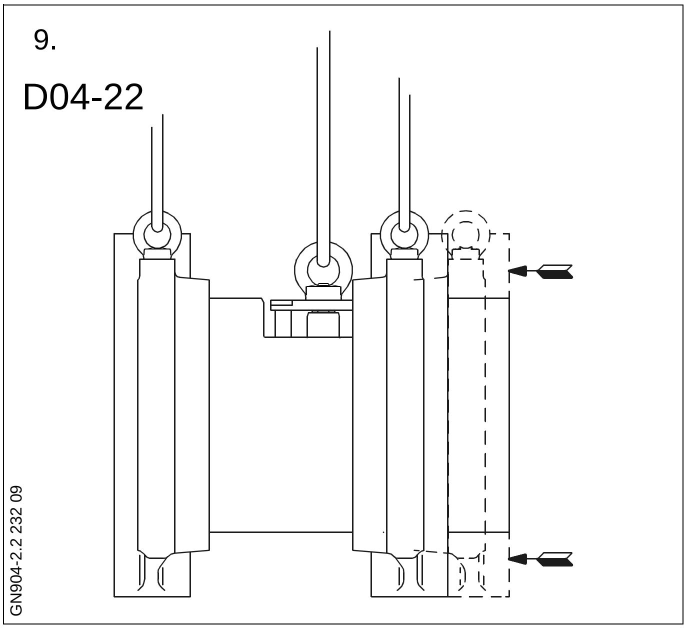
   Push one of the guide shoes against the middle of the crosshead. If necessary, use the guide shoe extractor tool.

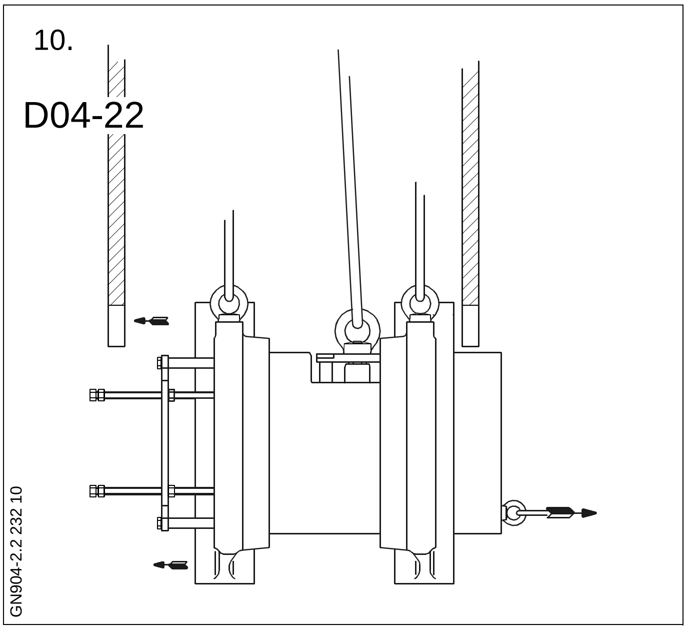
10. Pull the “free end” of the crosshead into the neighbouring cylinder unit.

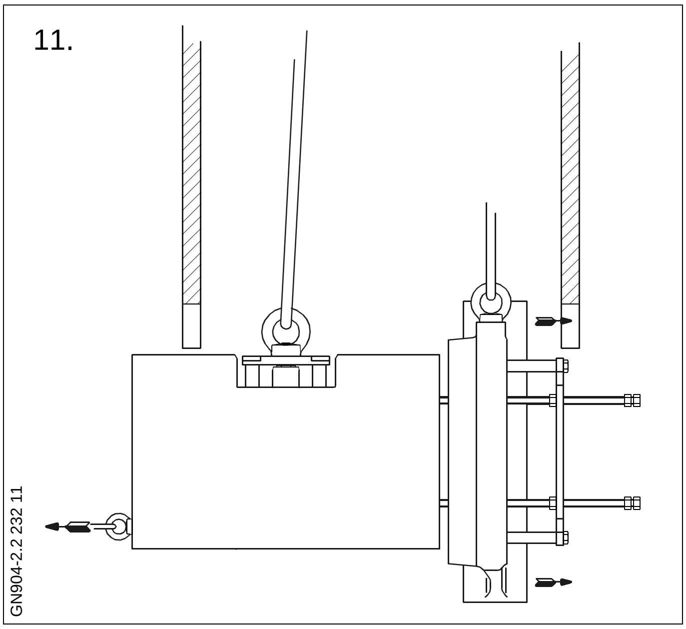
    Mount the guide shoe extractor on the opposite guide shoe.

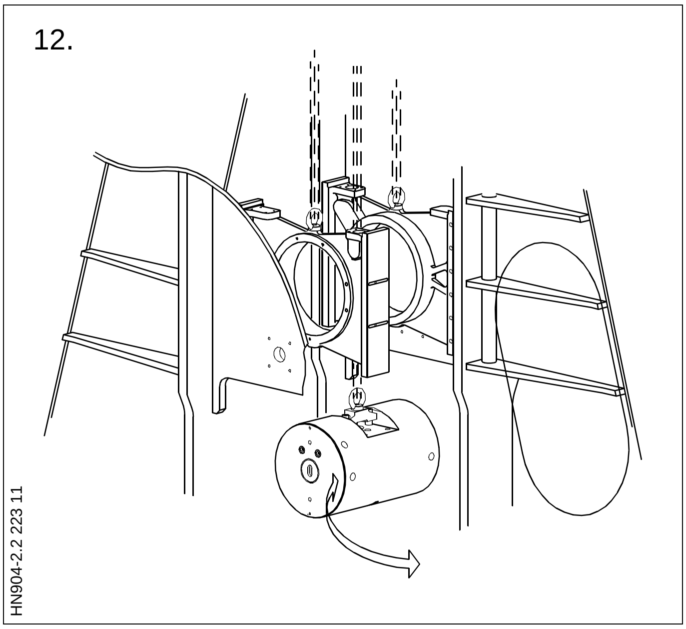
    Pull off the guide shoe, remove the extractor, and lift up the guide shoe.

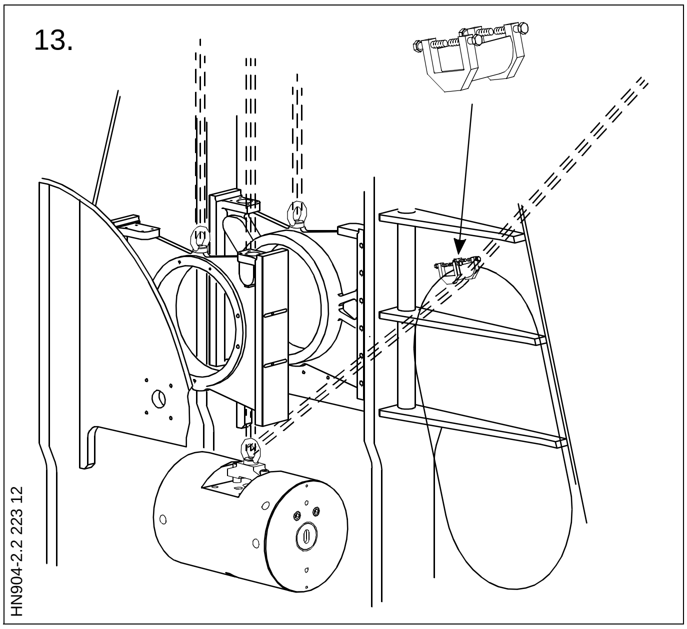












11. Pull the crosshead with guide shoe into the opposite neighbouring cylinder unit.

    Mount the guide shoe extractor tool on the remaining guide shoe.

    Pull off the guide shoe, remove the extractor tool and lift up the guide shoe.

12. Turn the crosshead 90°.

13. Mount the wire guide tool in the framebox door opening.

    Remove the crosshead from the engine, using wire rope and tackles.

    Protect the crosshead, for instance with thick or corrugated paper, and land it outside the engine.

    If necessary, remove the guide shoes from the engine.

1. Attach the flat-plaited wire strap to the engine room crane.

   Lift the crosshead into the engine, using wire rope, tackles, and the engine room crane.

2. Turn the crosshead 90°. Pull the crosshead to one side under the web plates between the cyliners.

3. Lower one guide shoe in the other side of the crosshead and mount the guide shoe extractor tool.

   Use the guide shoe extractor tool to pull the crosshead so far through the guide shoe that the other guide shoe can be lowered.

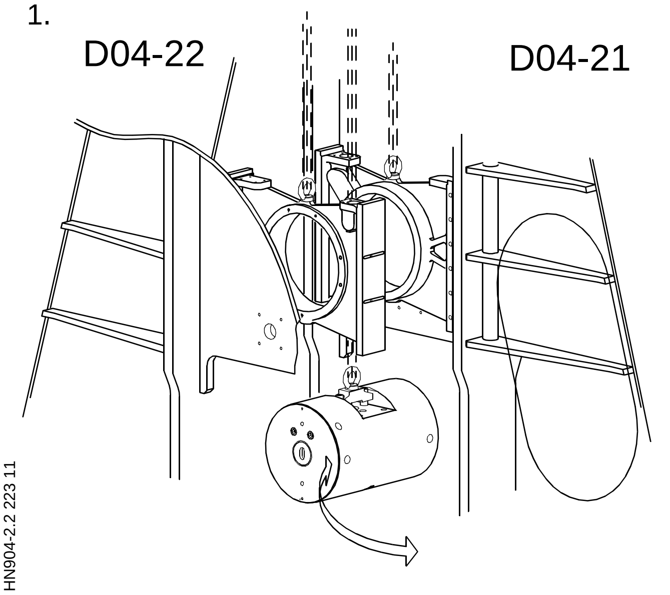
   Follow the movement of the crosshead with the engine room crane.

4. Lower the other guide shoe.

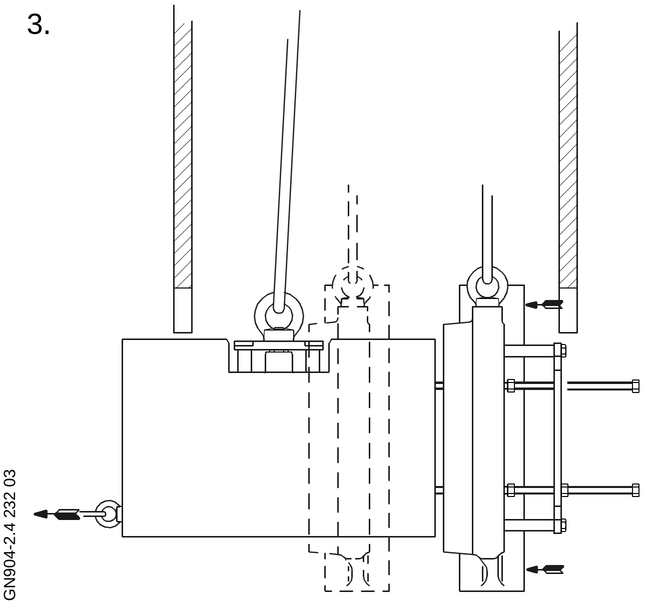
   Use the guide shoe extractor tool to push the crosshead towards the second guide shoe.

   Remove the extractor tool and adjust the position of the opposite guide shoe.

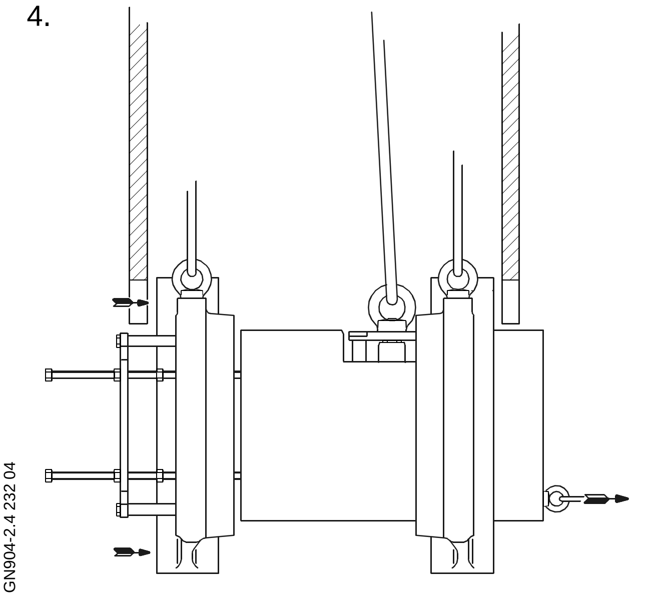
1.



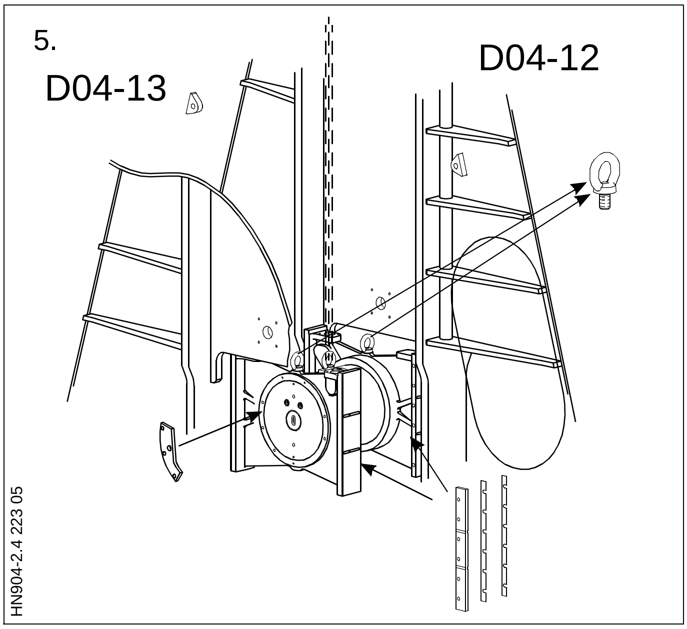
3.



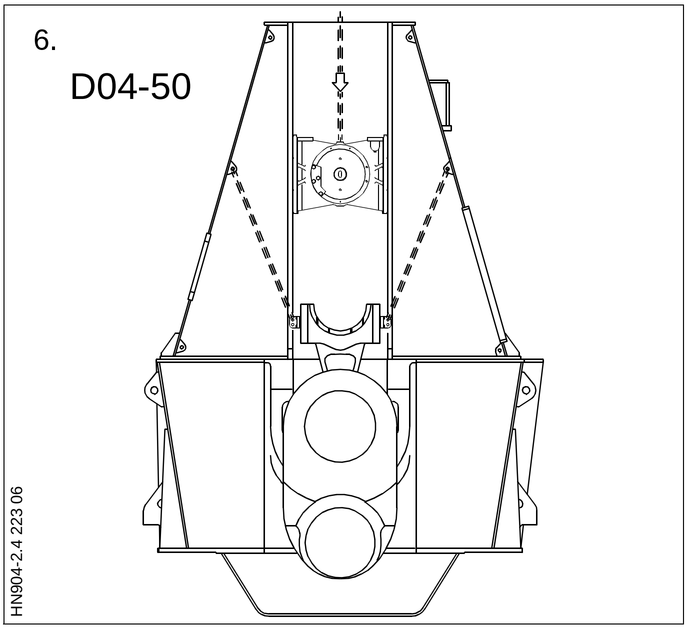
4.



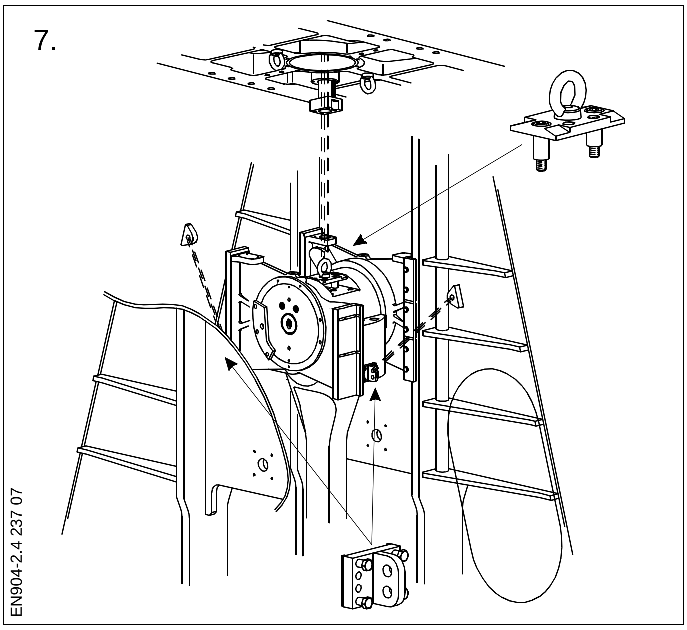
5.



6.



7.



5. Mount the guide plates on the side of the guide shoes.

   Tighten the screws and lock them with locking wire.
   See Procedure 913-7.

   Using the engine room crane, lift the crosshead to a working position and mount the guide strips on the side of the guide shoes.

   Tighten the screws and lock them with locking wire.
   See Procedure 913-7.

6. Using the tackles, lift the connecting rod to a vertical position while turning the crankthrow to BDC.

7. When the connecting rod is in a vertical position, lower the crosshead and land it on the connecting rod.

   Remove the lifting attachments from the connecting rod head.

   Remove the lifting tool from the crosshead.

8. Lift the crosshead bearing cap into the engine.

9. Mount the spacer rings and the hydraulic jacks for tightening the nuts on the crosshead bearing studs.

   For operation of the hydraulic tools, see Procedure 913.

   Mount the drain oil slotted pipe and the cooling oil outlet pipe on the guide shoe.

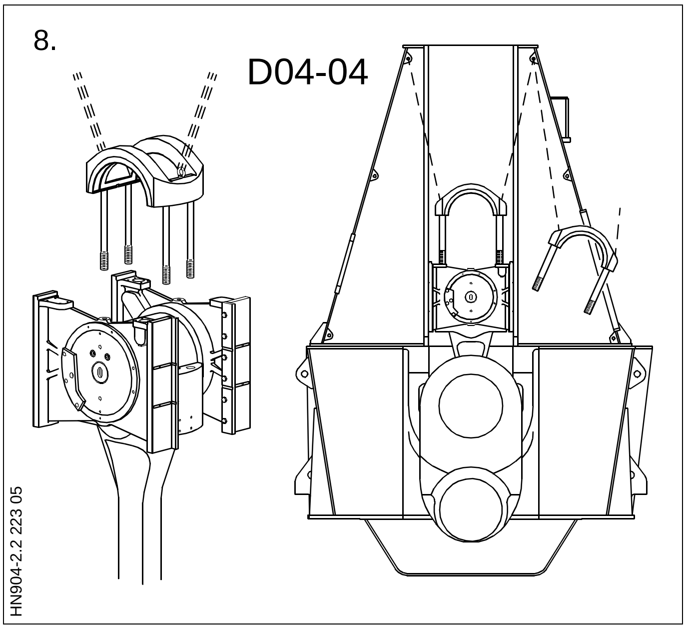
   Turn the crankshaft to TDC.

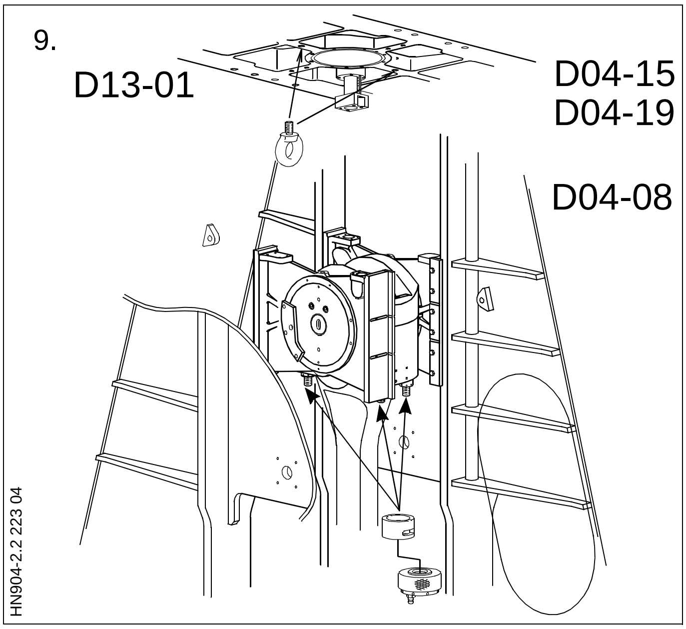
   Land the telescopic pipe on the guide shoe.

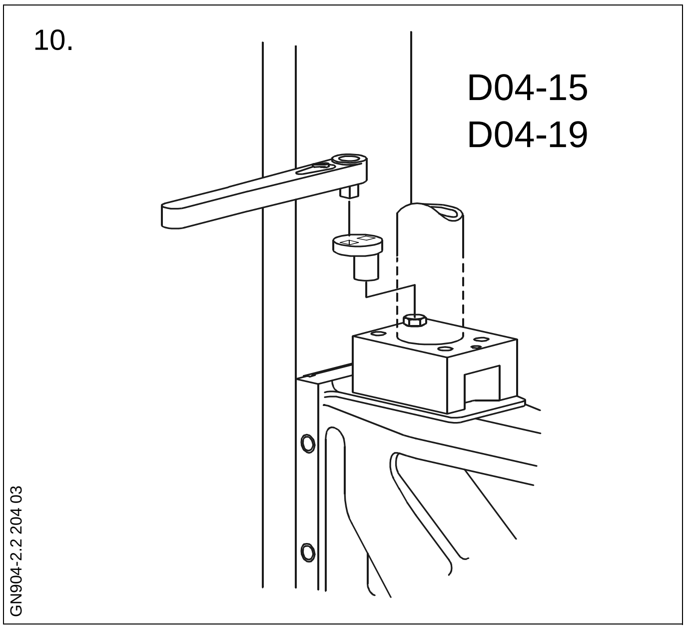
10. For tightening the bolt in the corner behind the telescopic pipe, use the offset tool along with the torque wrench.

11. Mount the lubricating oil pipes on the main bearing caps.

12. Mount the piston.
    See Procedure 902-1.4.







SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

X	Stopped engine
X	Shut off starting air supply – *At starting air receiver*
X	Block the main starting valve
X	Shut off starting air distributor/distributing system supply
X	Shut off safety air supply – *Not ME engines*
X	Shut off control air supply
	Shut off air supply to exhaust valve – *Only with stopped lubricating oil pumps*
X	Engage turning gear
	Shut off cooling water
	Shut off fuel oil
X	Stop lubricating oil supply
X	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
	Acceptance criteria with piston in centre (F-A direction)		
D04-25	PF+PA, N max.	0.5	mm
D04-26	PF+PA, O max.	0.7	mm
D04-28	E+G, H+F, N max.	0.5	mm
D04-29	E+G, H+F, N min.	0.2	mm
D04-30	E+G, H+F, O max.	0.8	mm
D04-34	J+C, L+D, K+C, M+D, N max.	0.9	mm
D04-35	J+C, L+D, K+C, M+D, N min.	0.5	mm
D04-36	J+C, L+D, K+C, M+D, O max.	1.1	mm
D04-40	ZF/ZA O max.	5.0	mm
D04-41	ZF/ZA O min.	4.0	mm
D04-67	Crankshaft position (after BDC).	70	°
D04-68	Piston inclination R1-R2, O max.	0.35	mm
	N: New and cold engine with staybolts tightened (less than 100 running hours).		
	O: Engine in service.		

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P91366	058	Feeler gauge
P91366	073	Dial gauge and stand tool

In order to achieve uniform measuring conditions on board, the ship's trim must be as close as possible to 0°.

1. Mount a transparent plastic tube along the length of the bedplate.

   Bend each end approx. 250 mm up the framebox side. *See* **T**.

   Fill the tube with water (preferably coloured) until the water level is approx. 100 mm from the end of the tube.

   Trim the ship until the difference between the water level **S** fore and aft is less than 1.5 mm per 1000 mm.

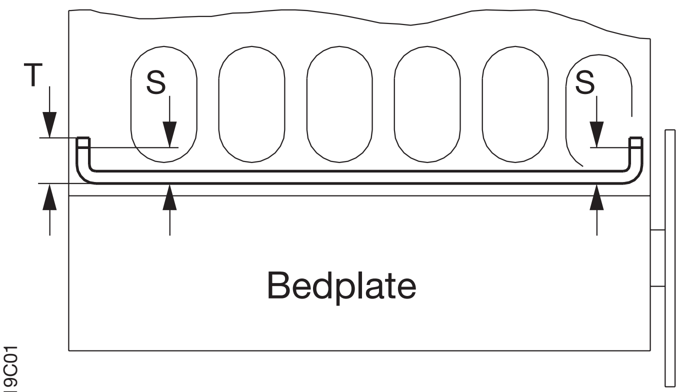
   Measurements are to be taken with a ruler.

2. Turn the crankshaft in ASTERN direction to D04-67 (see data) degrees after BDC (the guide shoe must rest against the crosshead guide).

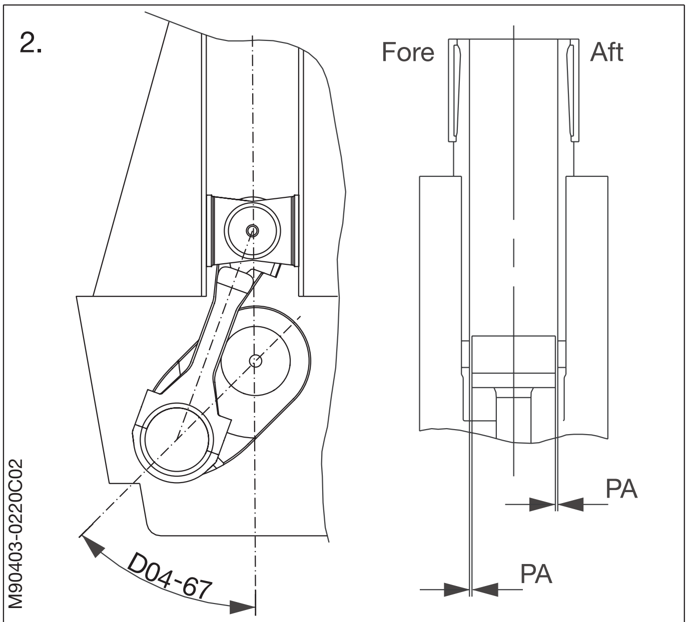
   Check the centering of the piston in the cylinder liner by measuring the clearance (use a long feeler gauge from the scavenge air space) between the piston skirt and the cylinder liner in the Fore and Aft positions (**PF-PA**).

   Make sure that the piston is clear of the cylinder liner in the fore and aft directions.

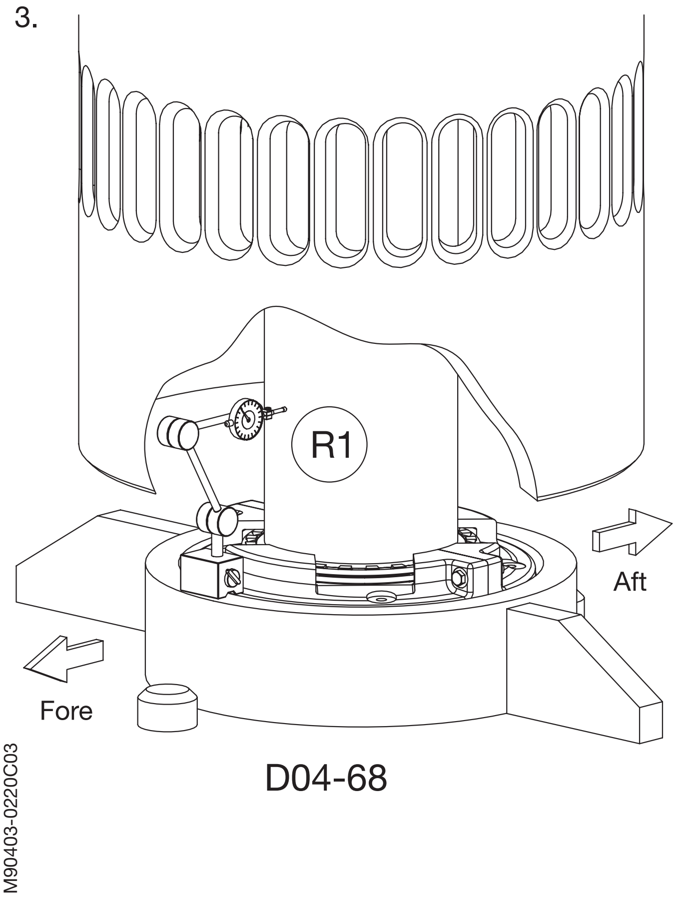
1.



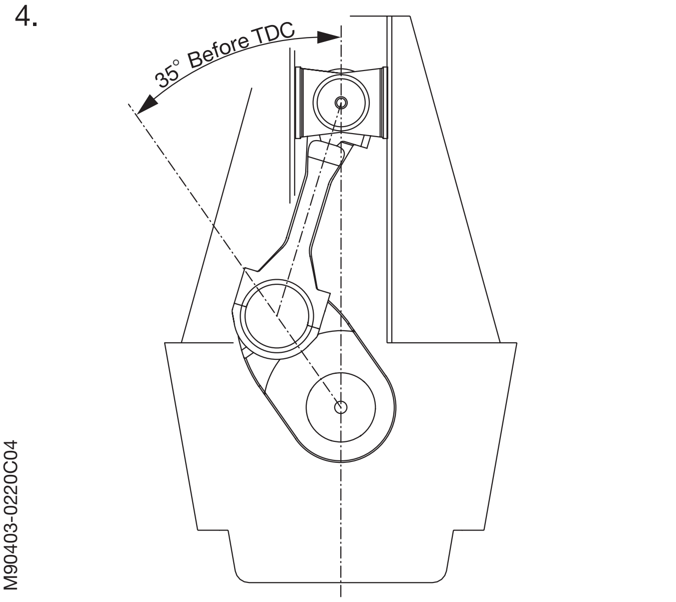
2.



3.



4.



3. Fit a dial gauge in the bottom of the scavenge air space to the front of the piston rod with the tip of the dial gauge against the piston rod.

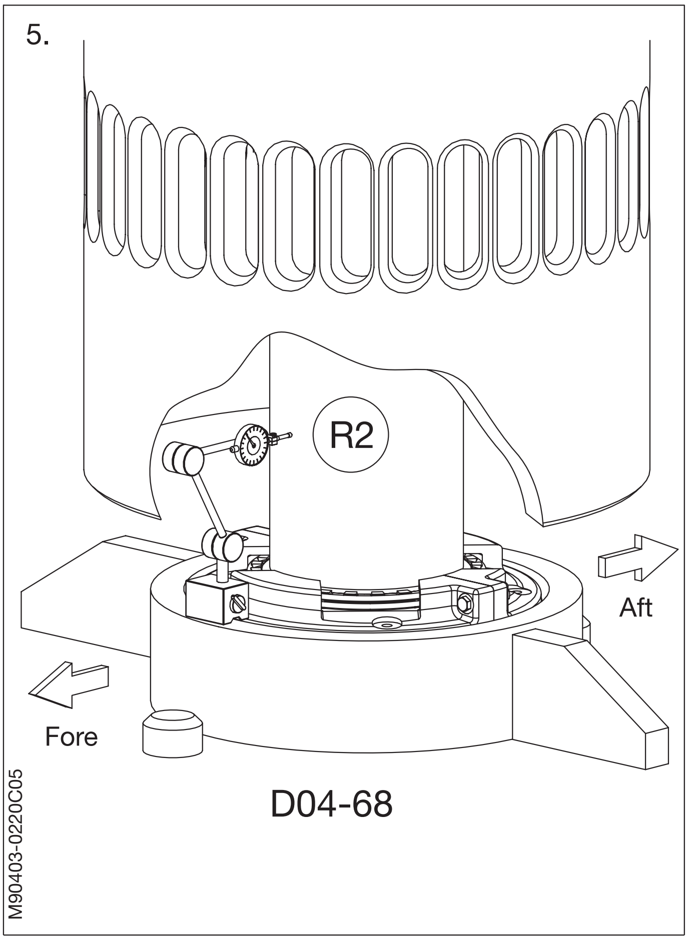
   Note down the reading of the dial gauge R1.

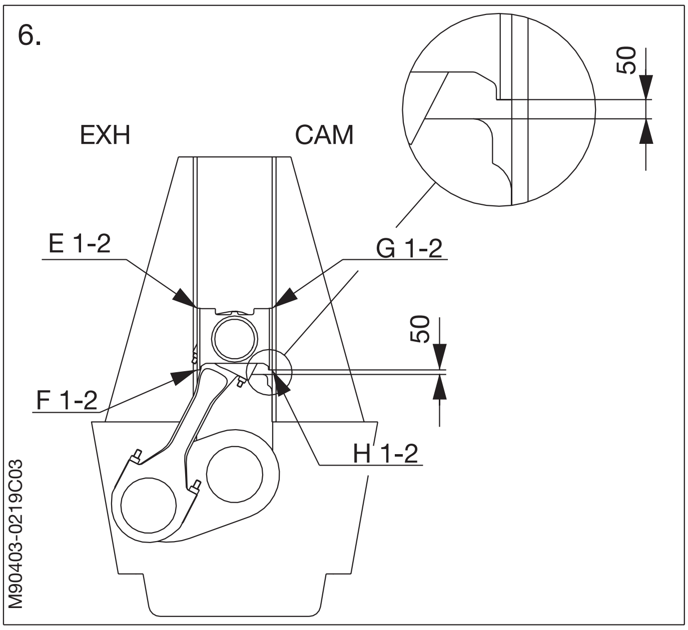
4. Turn the crankshaft in ASTERN direction to 35° before TDC.

5. Note down the reading of the dial gauge R2.

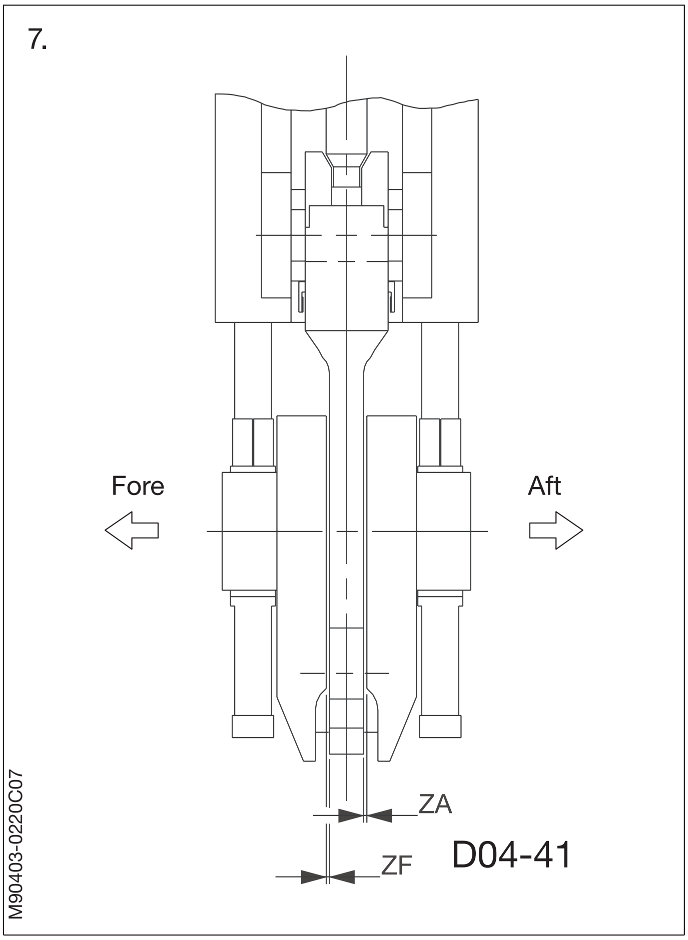
   Calculate the piston inclination R1-R2.

6. Turn in astern direction until the bottom of the guide shoe is 50 mm above the top of the cut-out in the web plate.

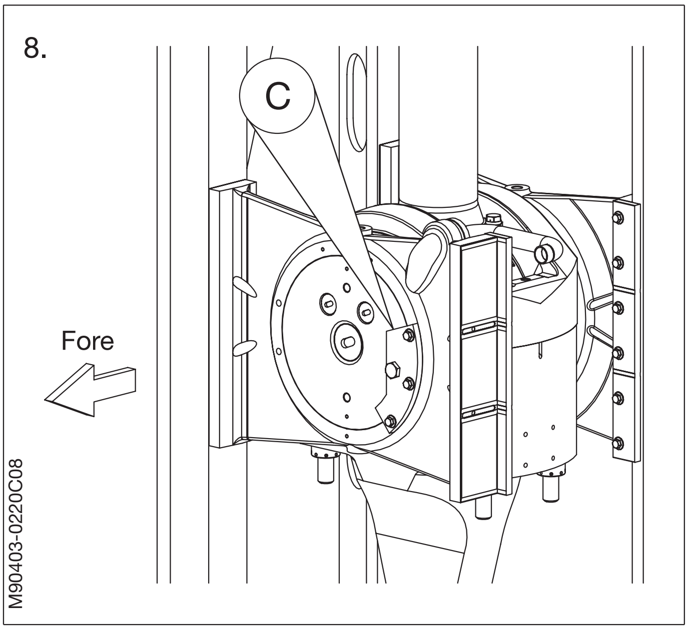




7.



8.



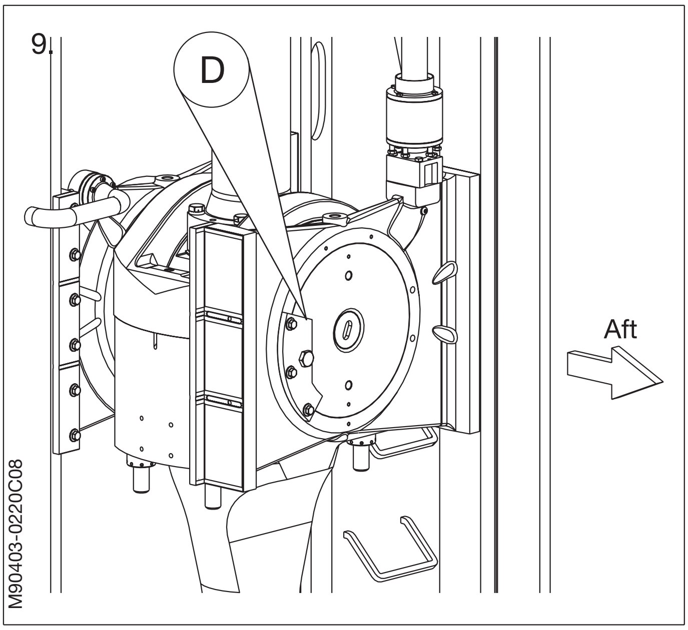
7. Measure the fore and aft clearances ZF and ZA between the crankthrow and the connecting rod.

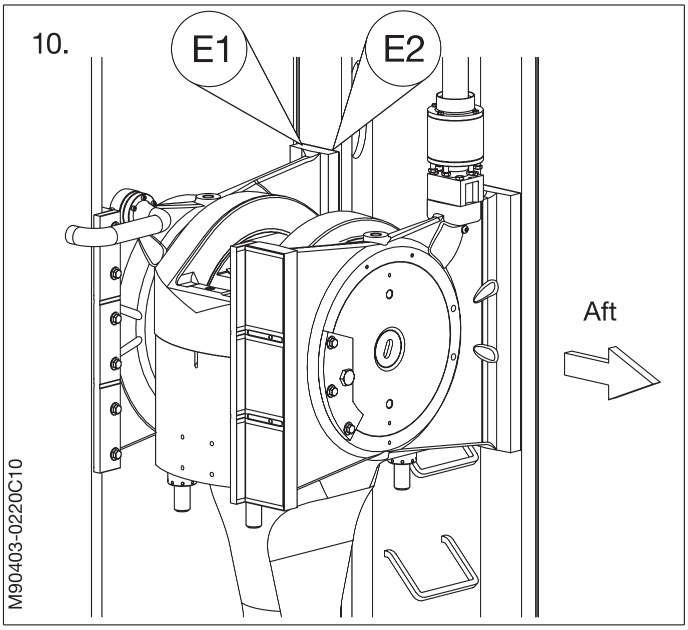
8. Measure the clearance C, between the foremost stop plate and the crosshead.

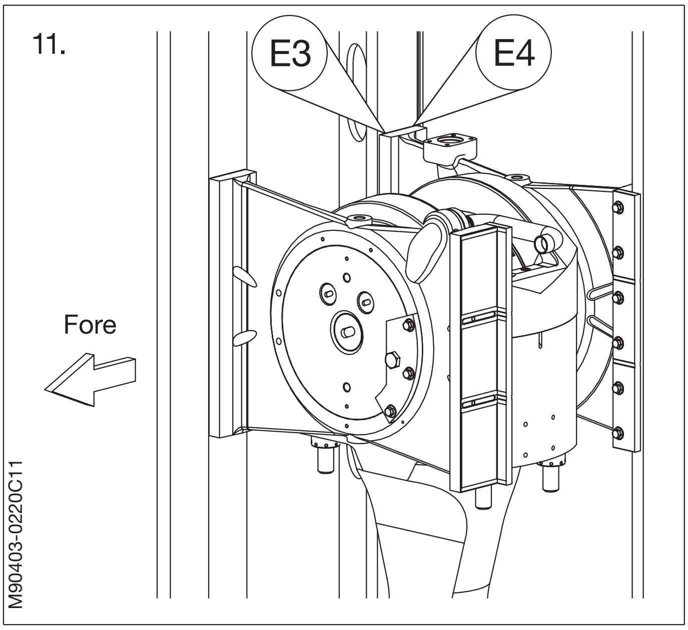
9. Measure the clearance D, between the aftmost stop plate and the crosshead.

10. Measure the clearances E1 and E2 between the top of the foremost guide shoe and the exhaust-side crosshead guide.

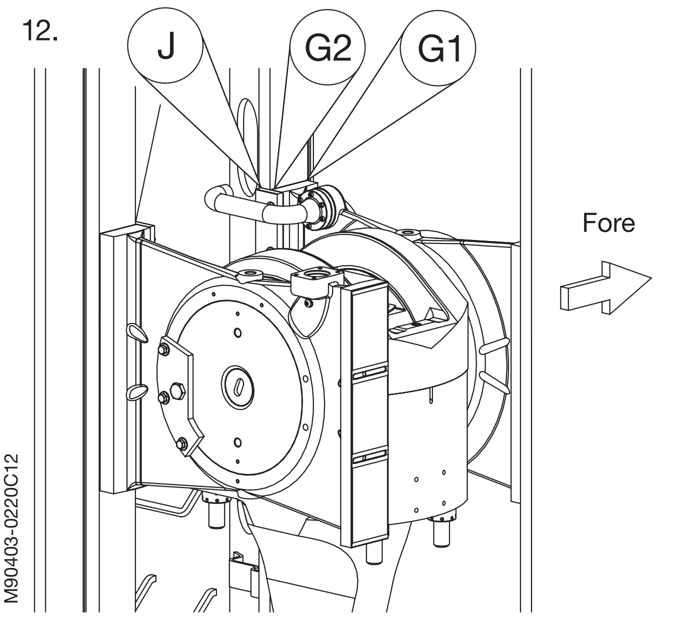
11. Measure the clearances E3 and E4 between the top of the aftmost guide shoe and the exhaust-side crosshead guide.



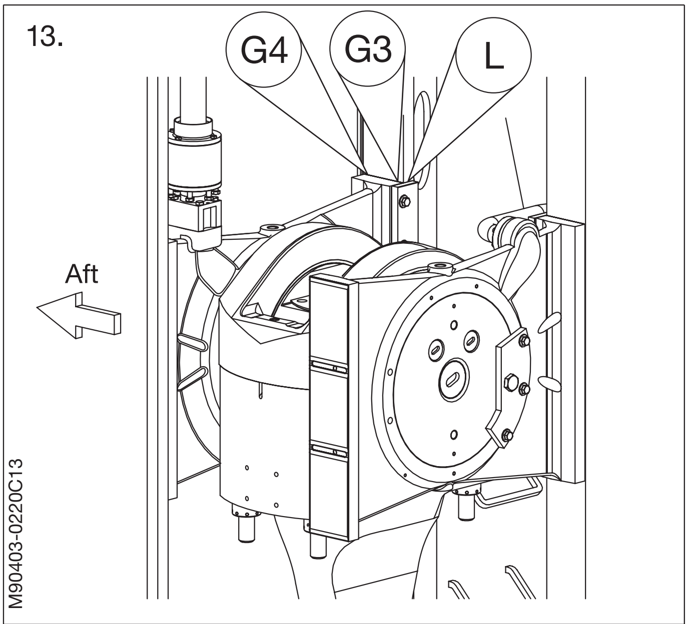




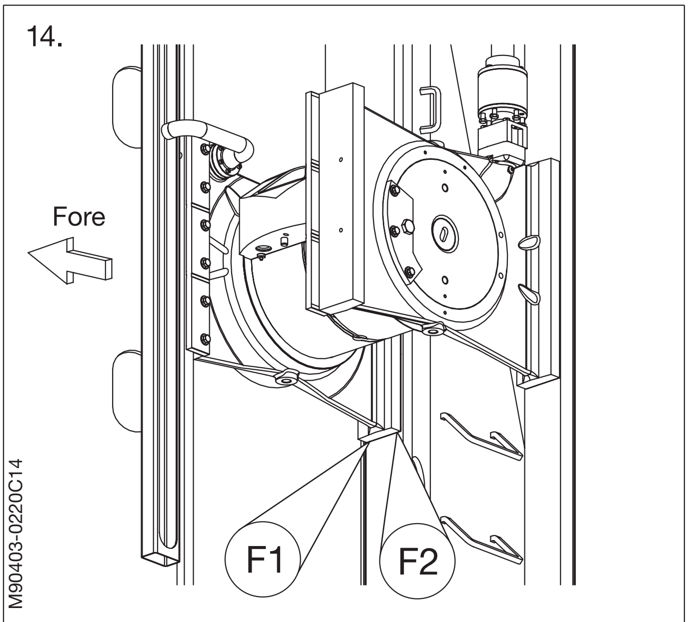
12.



13.



14.



12. Measure the clearances G1 and G2 between the top of the foremost guide shoe and the manoeuvring-side crosshead guide.

    Measure the clearance J between the top of the foremost guide strip and the manoeuvring-side crosshead guide.

13. Measure the clearances G3 and G4 between the top of the aftmost guide shoe and the manoeuvring-side crosshead guide.

    Measure the clearance L between the top of the aftmost guide strip and the manoeuvring-side crosshead guide.

14. Measure the clearances F1 and F2 between the bottom of the foremost guide shoe and the exhaust-side crosshead guide.

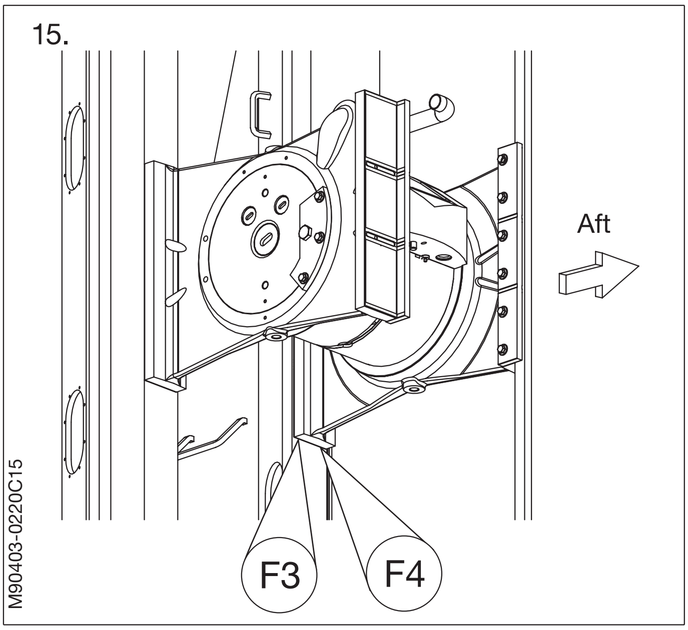
15. Measure the clearances F3 and F4 between the bottom of the aftmost guide shoe and the exhaust-side crosshead guide.

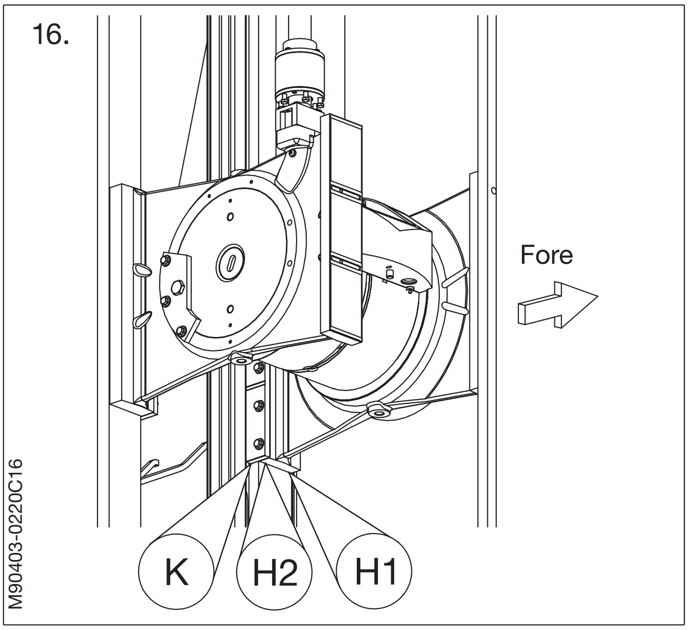
16. Measure the clearances H1 and H2 between the bottom of the foremost guide shoe and the manoeuvring-side crosshead guide.

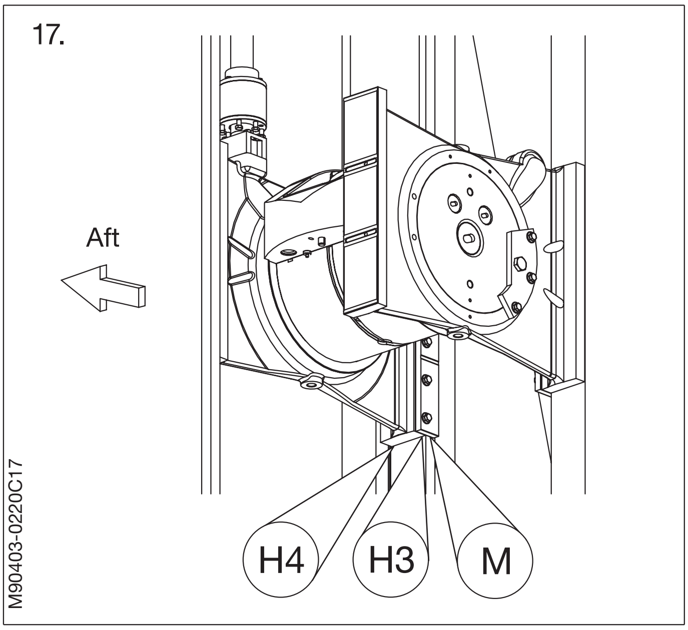
    Measure the clearance K between the bottom of the foremost guide strip and the manoeuvring-side crosshead guide.

17. Measure the clearances H3 and H4 between the bottom of the aftmost guide shoe and the manoeuvring-side crosshead guide.

    Measure the clearance M between the bottom of the aftmost guide strip and the manoeuvring-side crosshead guide.







19.

Unit mm	Cyl.	1	2	3
Piston/Liner	PF			
	PA			
Framebox	E1			
	E2			
	E3			
	E4			
	F1			
	F2			
	F3			
	F4			
	G1			
	G2			
	G3			
	G4			
	H1			
	H2			
	H3			
	H4			
	S			
	K			
	L			
	M			
	C			
	D			
Piston Rod	R1			
	R2			
Crankthrow	ZF			
	ZA			

M90403-0220C19

18. The guide strip clearance, calculated as **J+C**, **K+C**, **L+D** and **M+D**, is adjusted by the insertion of shims so that it is symmetrical in relation to the clearance between the piston skirt and the cylinder liner.

    Parallelism between the guide strip and guide is to be kept within a tolerance of 0.2 mm per 1000 mm.

19. It is recommended that the measured results are noted down so that possible later changes can be ascertained.

    Compare the measured results with the values stated on the data sheet.

SAFETY PRECAUTIONS

X	Stopped engine
X	Block the starting mechanism
X	Shut off starting air supply
X	Engage turning gear
	Shut off cooling water
	Shut off fuel oil
X	Shut off lubricating oil
	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D04-43	Crankpin bearing clearance, max.	0.6	mm
D04-44	Crankpin bearing clearance, min.	0.2	mm
D04-46	Crankpin bearing, upper shell	17	kg
D04-47	Crankpin bearing, lower shell	13	kg
D04-48	Crankpin bearing cap + shell + bearing studs	150	kg
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400 -1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P**909**51 refers to chapter 909.

Plate	Item No.	Description
P90451	47	Wire guide
P90451	59	Lifting attachment - connecting rod
P90451	96	Bracket for support, crosshead
P90451	106	Bracket for support, crosshead
P90461		Connecting Rod - Hydraulic Tools
P91356		Lifting Tools, Etc.
P91366	50	Feeler gauge
P91351	10	Hydraulic pump, pneumatically operated
P91351	46	Hose with unions (1500 mm), complete
P91351	58	Hose with unions (3000 mm), complete
P91351	105	3-way distributor block, complete

The bottom clearance between the journal and a new bearing shell is the result of a summation of the production tolerances of the bearing assembly components.

For the bottom clearance of a specific bearing, see the measurement in the Adjustment Sheet in Volume I, OPERATION.

1. Open the crankcase door at the relevant cylinder.

2. Turn the crank concerned to BDC.

3. Measure the clearance in the crankpin bearing by inserting a feeler gauge at the bottom of the bearing shell in both sides. *See Data for bottom clearance.*

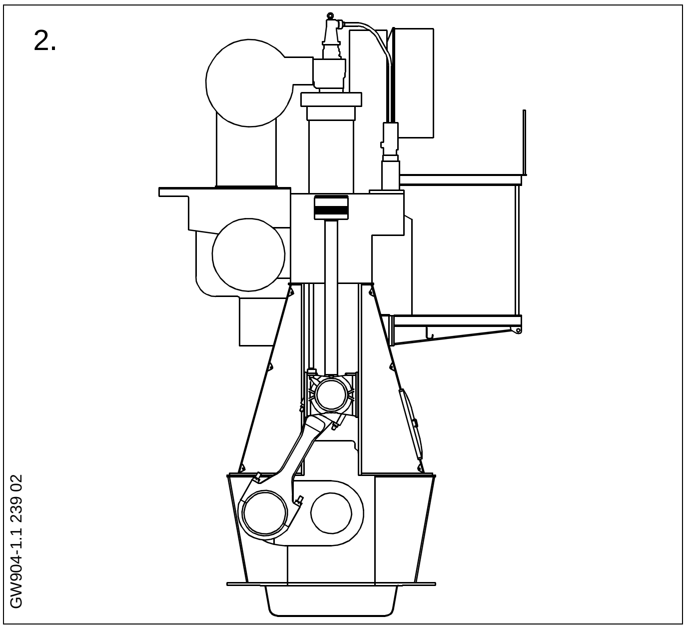
4. The difference between the **actual** clearance measurement and the measurement recorded in the Adjustment Sheet (or the clearance noted for a new bearing installed later) **must not** exceed 0.1 mm. If so, the crankpin bearing must be disassembled for inspection.
See Procedure 904-4.2.

5. The wear limit for the crankpin bearing shells is based on an evaluation of the bearing condition at the time of inspection.

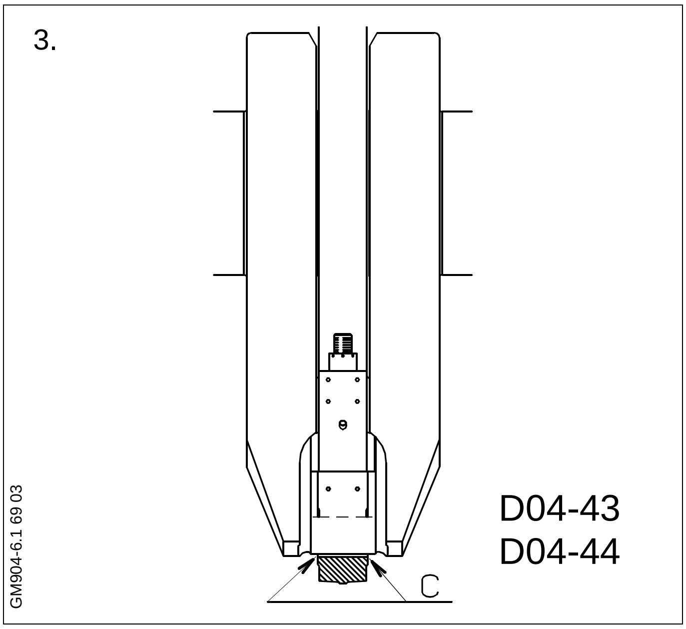
   An average wear rate of 0.01 mm per 10,000 hours is regarded as normal.

6. *For further external inspection of the crankpin bearing, see Chapter 708 ‘Bearings' in the instruction book, Volume I, OPERATION.*

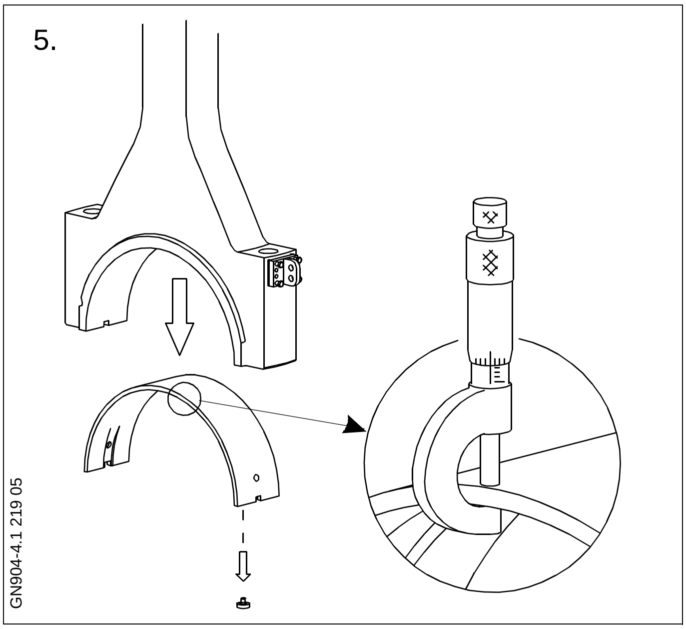
2.



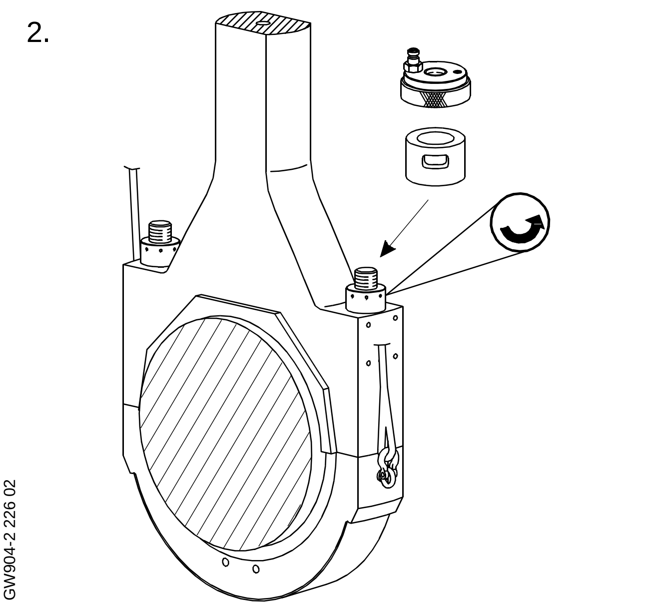
3.



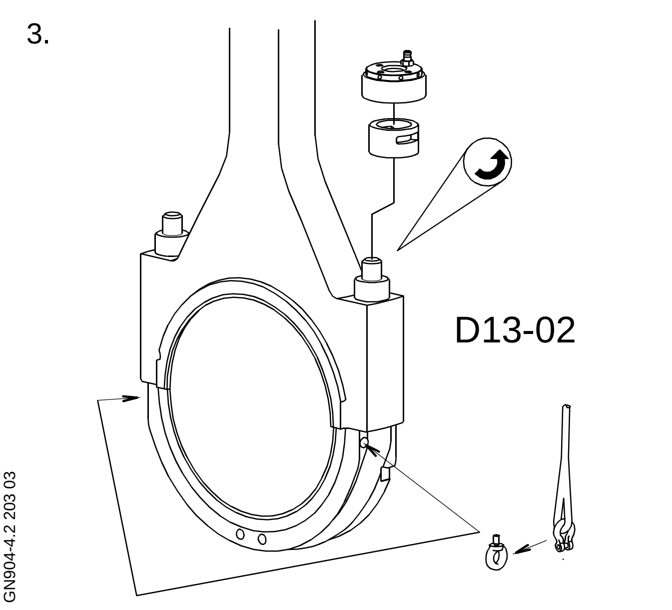
5.



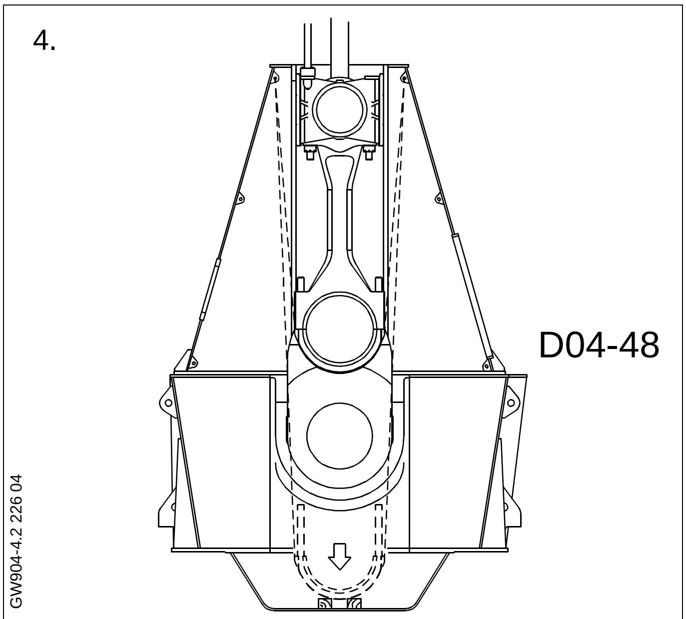
2.



3.



4.



1. Turn the crank to BDC.

2. Suspend two tackles from the lifting brackets, in the athwartship direction.

3. Turn the crank to TDC.

   Mount eye bolts in each side of the crankpin bearing cap and, using shackles and wire ropes, hook on the tackles and haul tight.

   Loosen the crankpin bearing stud nuts, using the hydraulic jacks.

   For operation of the hydraulic jacks, see Procedure 913-1.

   Remove the hydraulic jacks and the nuts.

4. Lower the bearing cap while seeing **carefully** that the studs do not damage the crankpin journal.

   Land the bearing cap on a couple of planks placed in the oil pan.

   Inspect the bearing shell.

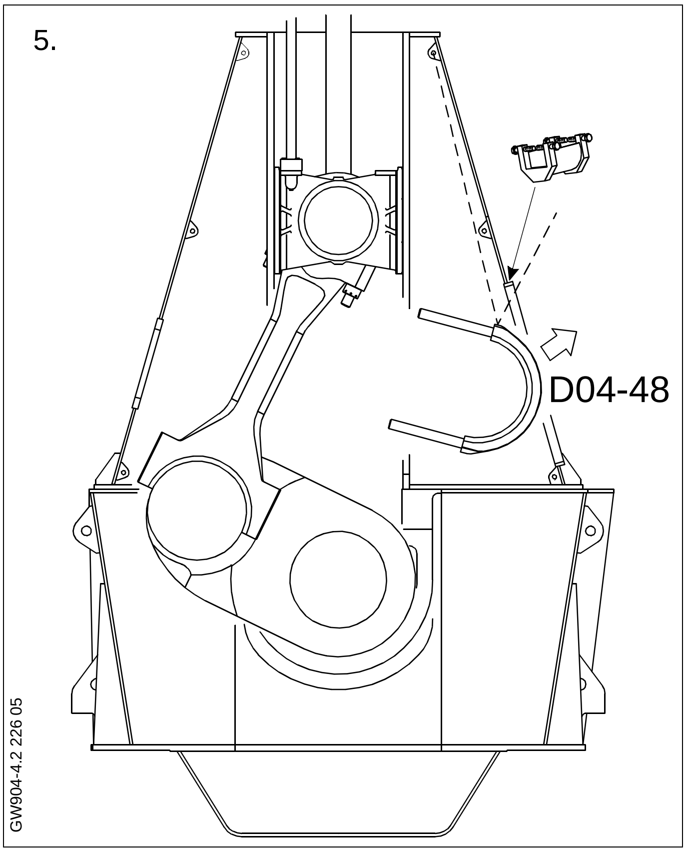
   For evaluation of bearings, see Volume I, Chapter 708.

5. If the bearing shell needs to be replaced, remove the whole bearing cap from the crankcase.

   Hook the tackle on to an eye bolt on one side of the bearing cap.

   Mount the wire guide in the top of the crankcase door opening.

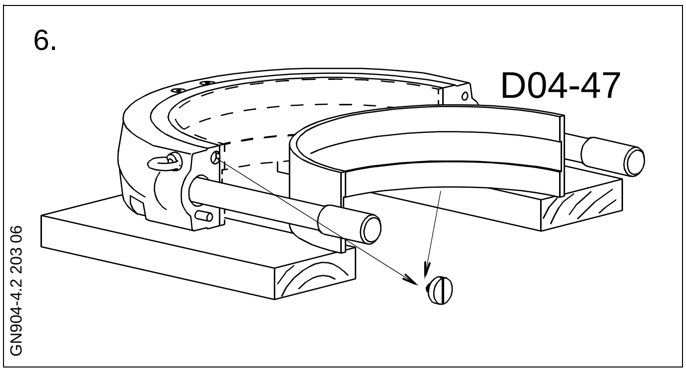
   Using the tackle from the frame box inside wall, together with a tackle suspended from the platform bracket, lift the bearing cap out of the crankcase.



6. Place the bearing cap on one side on a couple of planks.

   Inspect the bearing shell.
   See Volume I, Chapter 708.

   If necessary, dismount the bearing shell lock screws and replace the bearing shell by a new one.



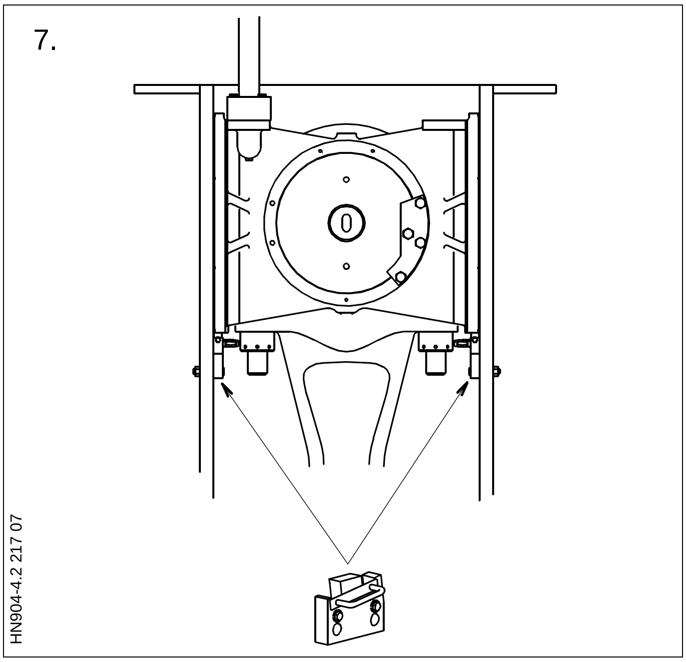
Note!
Normally bearing shells are replaced in pairs. If only one of the shells needs replacement, MAN B&W Diesel should be contacted for advice beforehand.

7. Turn to TDC.

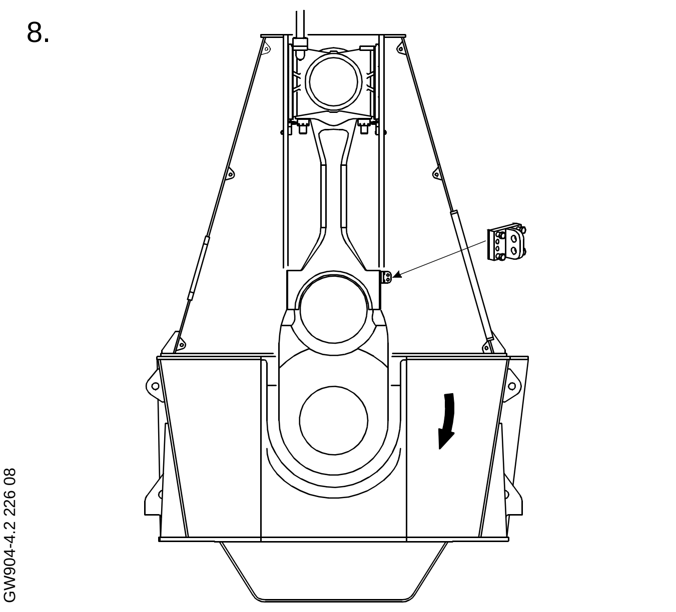
   Mount the four supports for guide shoes on the crosshead guides.

   Carefully turn the crank down until the guide shoes rest on the supports.

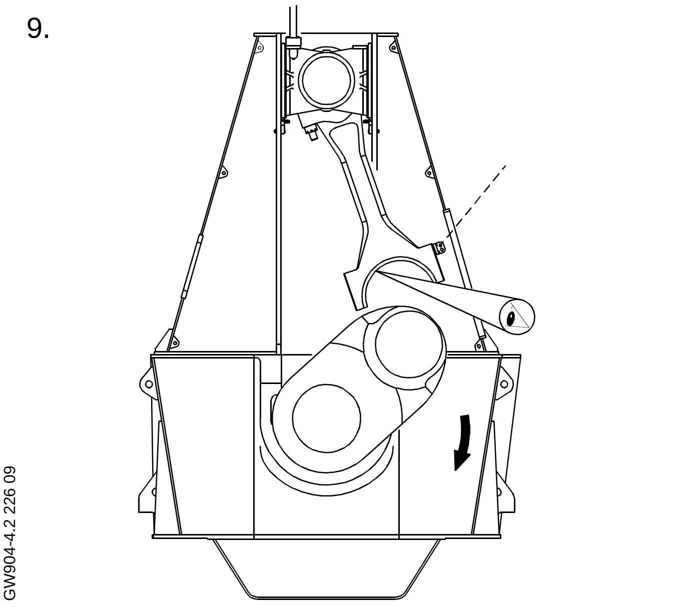
   Adjust the support brackets to the guide shoes so that the weight of the crosshead is evenly distributed on the four supports.



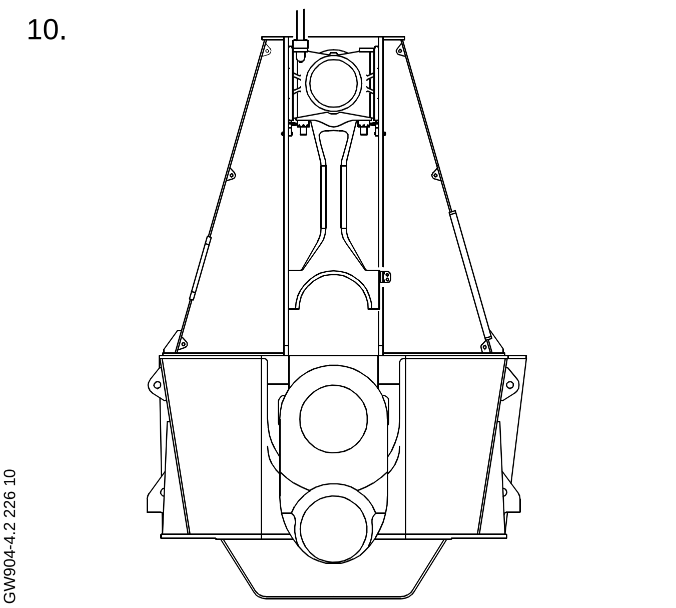
8.



9.



10.



8. Mount a lifting attachment for securing the connecting rod at the lower end, on one side.

   Hook on the tackle to a beam under the gallery platform and haul tight.

9. **Carefully** turn the crankshaft downwards, while ‘following’ with the tackle, making sure that the upper part of the bearing comes completely clear of the recess in the crankshaft when the parts begin to ‘separate’.

   Continue turning the crankshaft until the bearing surface can be freely inspected.

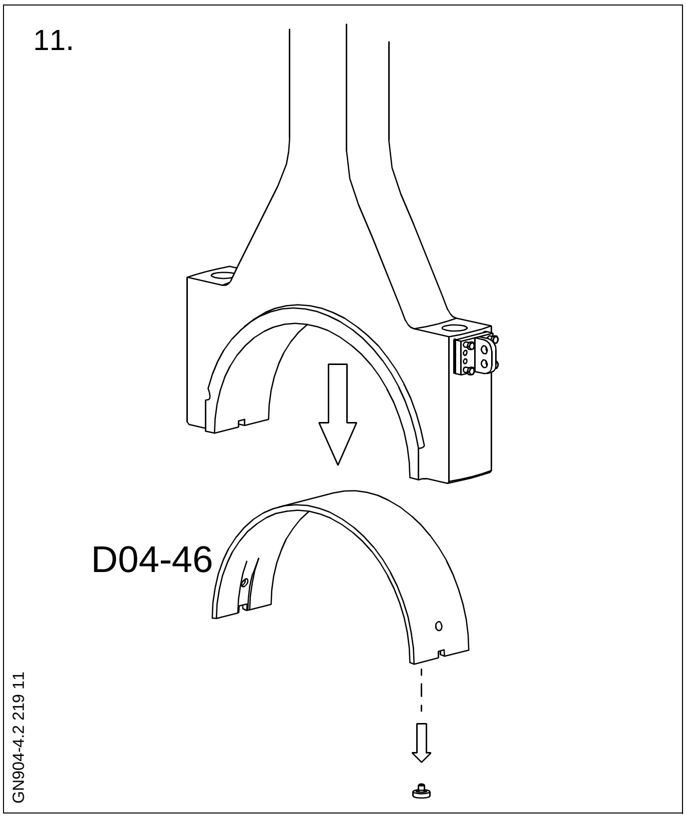
   Inspect the bearing shell surface and the crankpin journal.

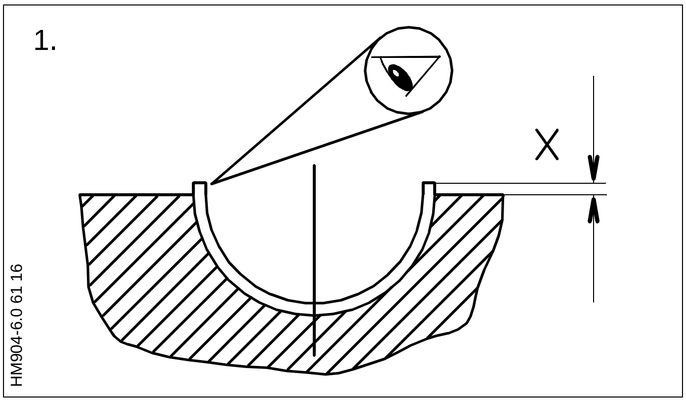
10. If it is necessary to replace the bearing shell, proceed as follows:

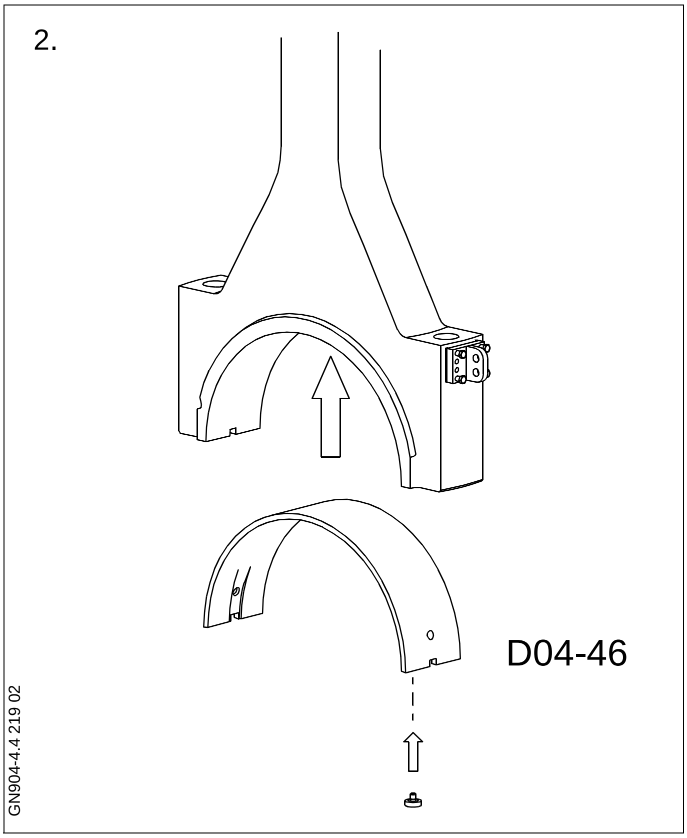
    Turn the crankshaft to BDC.

    Release the tackle so that the connecting rod is hanging freely.

11. Dismount the bearing shell lock screws and lift out the bearing shell.







1. Bearing shells of three mm undersize are available as spares in case of journal rectification. Please contact MAN B&W Diesel for advice.

   Coat the bearing shell surfaces and the journal with clean oil.

   The excess height **X** is to ensure the correct tightening-down of the bearing shell, and **must not** be eliminated.

2. Lift the upper bearing shell for the crankpin concerned into the crankcase.

   Carefully lift the bearing shell into position in the connecting rod, and mount the lock screws.

3. Hook the tackle on to a beam under the gallery platform and on to the lifting attachment on the connecting rod, and haul tight.

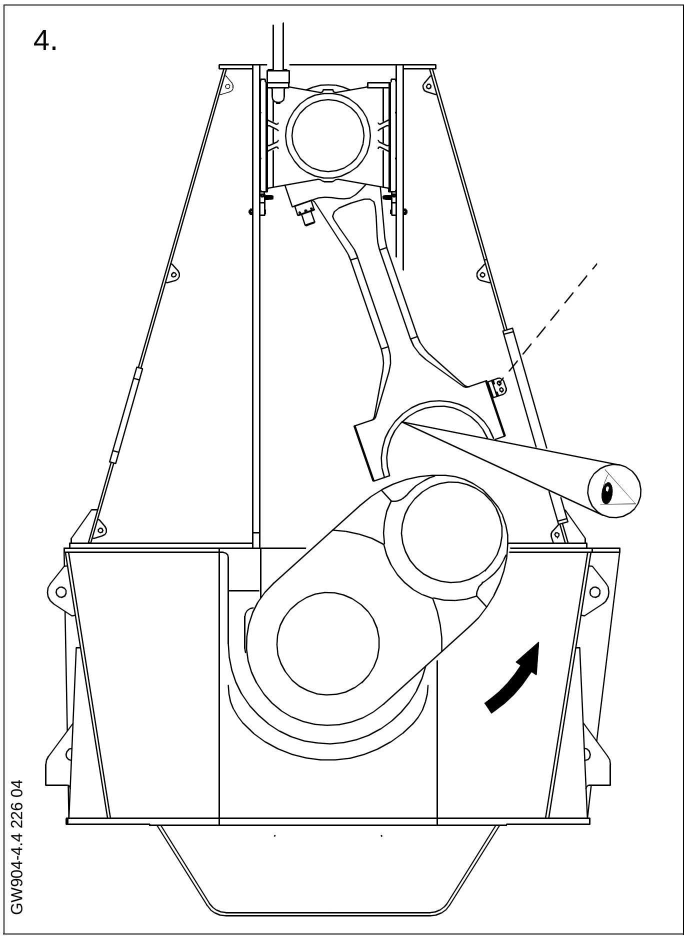
   Carefully turn the crankshaft upwards, while following up with the tackle, making sure that the upper part of the bearing enters the recess in the crankshaft when the parts turn together.

   Remove the tackle and the lifting attachment from the connecting rod.

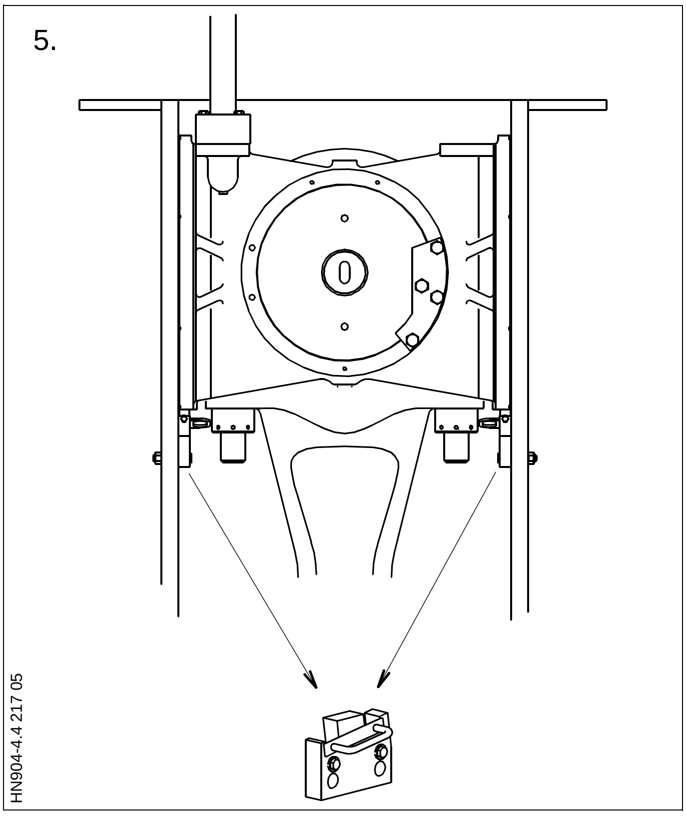
4. Turn the crosshead to TDC.

   Remove the guide shoe support brackets from the crosshead guides.

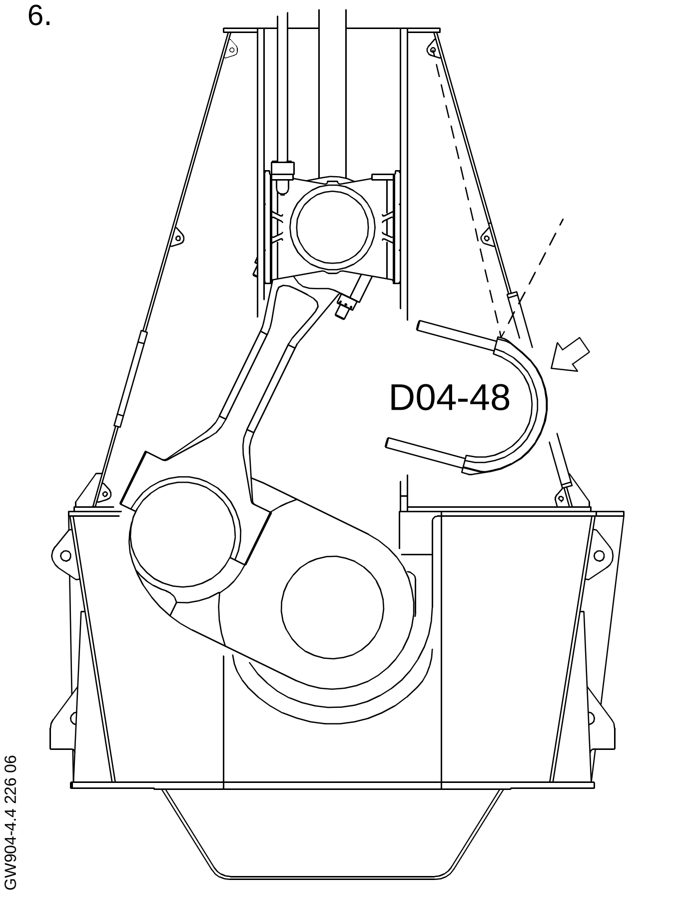
4.



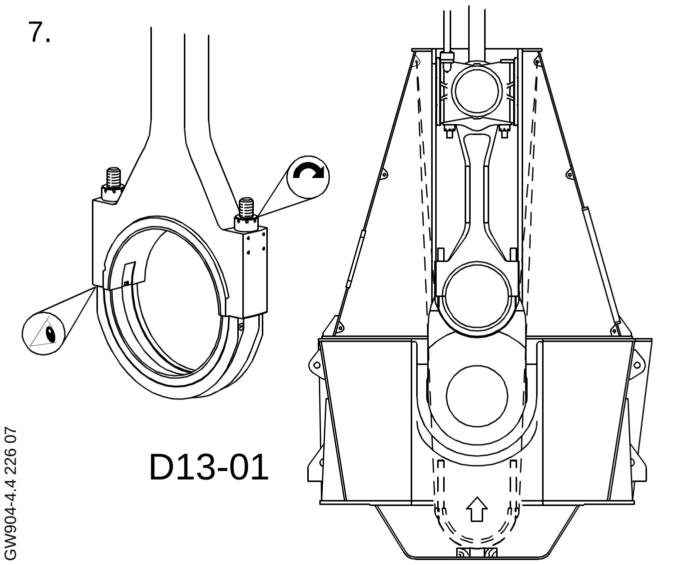
5.



6.



7.



5. Suspend the tackles from the lifting brackets in the top of the frame box.

   Lift the bearing cap assembly into the crankcase and land it on a couple of planks placed in the oil pan.

6. Hook the tackles on to the wire ropes and lift the bearing cap into position against the connecting rod.

Caution!
During mounting, take care that the studs do not damage the crankpin journal, and check that the guide pins mounted in the bearing cap enter the holes in the connecting rod.

Mount the nuts and, by means of spacer rings and hydraulic jacks, tighten the crankpin bearing cap. *See Data.*

For operation of hydraulic jacks, see Section 913.

Remove the tackles from the top of the frame box.

Check the bearing clearance.
See Procedure 904-4.1.

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

X	Stopped engine
X	Shut off starting air supply – *At starting air receiver*
X	Block the main starting valve
X	Shut off starting air distributor/distributing system supply
X	Shut off safety air supply – *Not ME engines*
X	Shut off control air supply
	Shut off air supply to exhaust valve – *Only with stopped lubricating oil pumps*
X	Engage turning gear
	Shut off cooling water
	Shut off fuel oil
X	Stop lubricating oil supply
	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D04-48	Crankpin bearing cap + shell + bearing studs	160	kg
D04-50	Connecting rod, without bearing caps	950	kg
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400-1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90451	047	Wire guide
P90451	059	Lifting attachment - connecting rod
P90451	060	Lifting tools - crosshead
P90451	096	Bracket for support, crosshead
P90451	106	Bracket for support, crosshead
P90461		Connecting rod - hydraulic tools
P91351	010	Hydraulic pump, pneumatically operated
P91351	046	Hose with unions (1500 mm), complete
P91351	058	Hose with unions (3000 mm), complete
P91351	105	3-way distributor block, complete
P91351	117	5-way distributor block, complete
P91356		Lifting tools, etc.
P91366	058	Feeler gauge

1. Turn the crank to BDC.

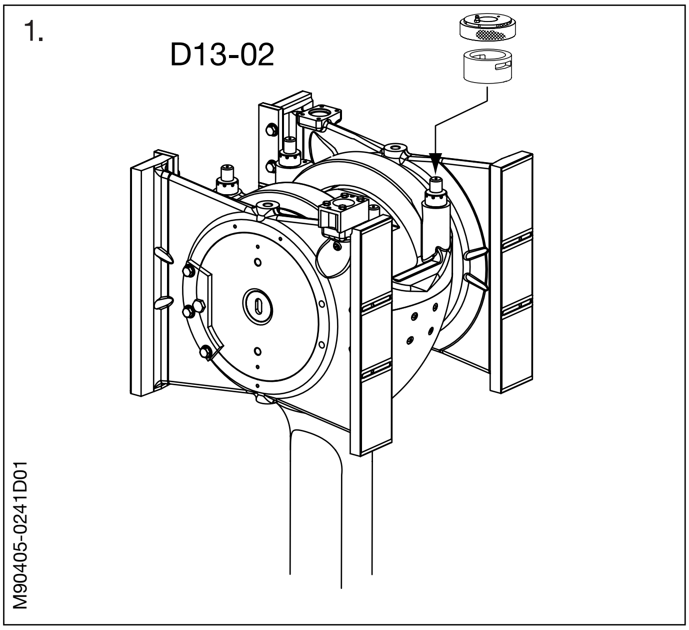
   Dismount the crosshead bearing stud nuts.
   See Procedure 904-1.2.

2. Mount two shackles in the top of the crankcase in the lifting brackets, in the athwartship direction, and suspend two tackles.

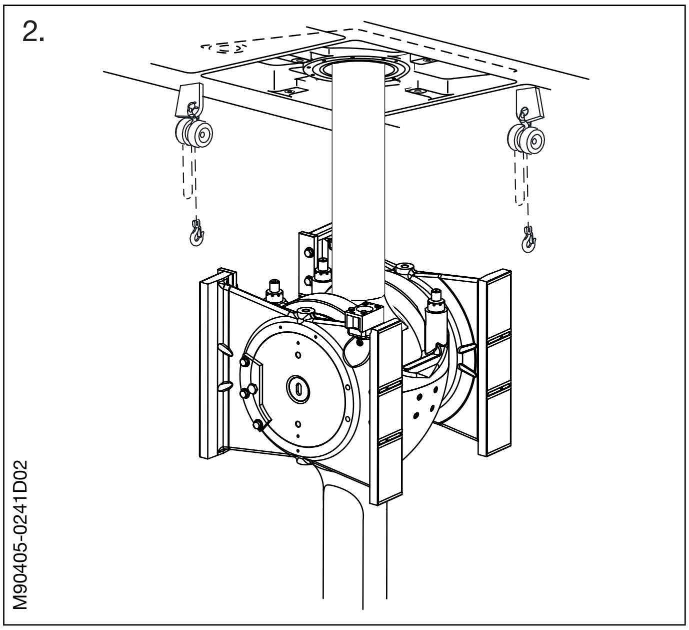
3. Turn the crank to TDC.

   Dismount the crankpin bearing cap, and remove the bearing cap from the engine.
   See Procedure 904-4.2.

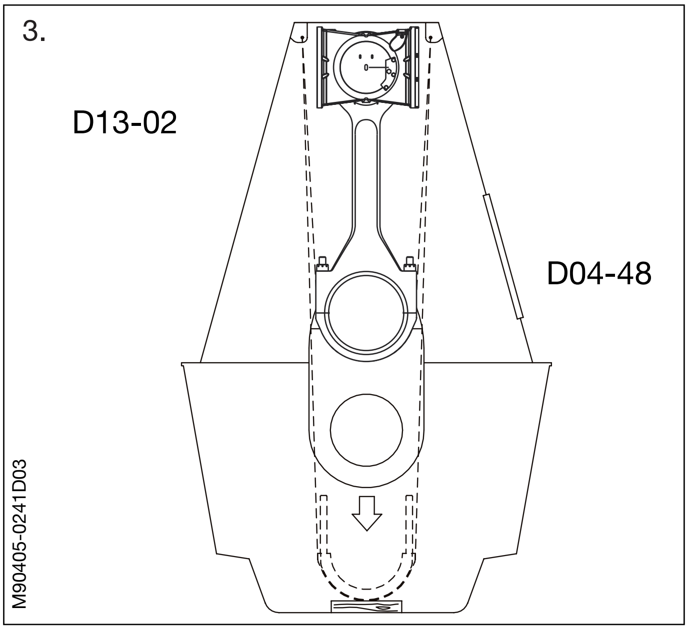
1.



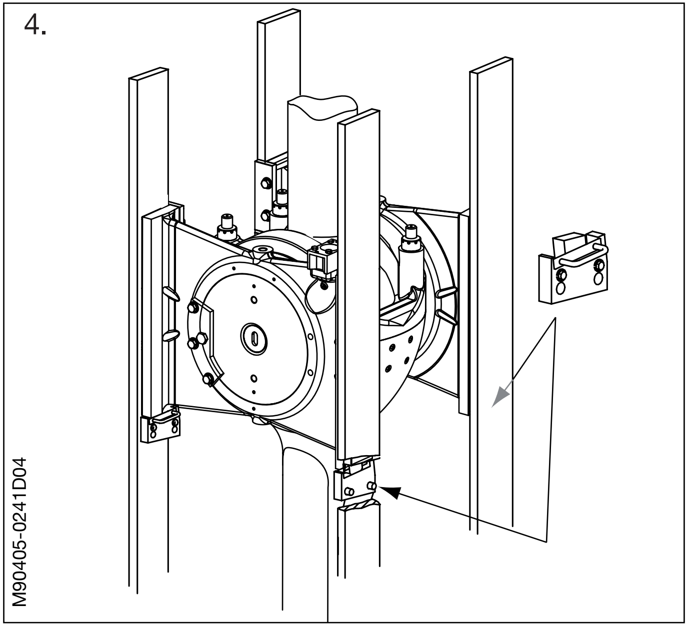
2.



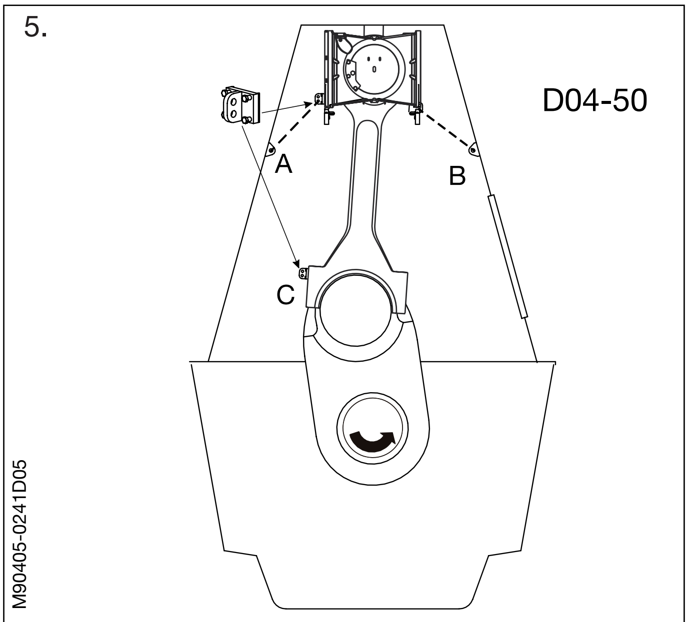
3.



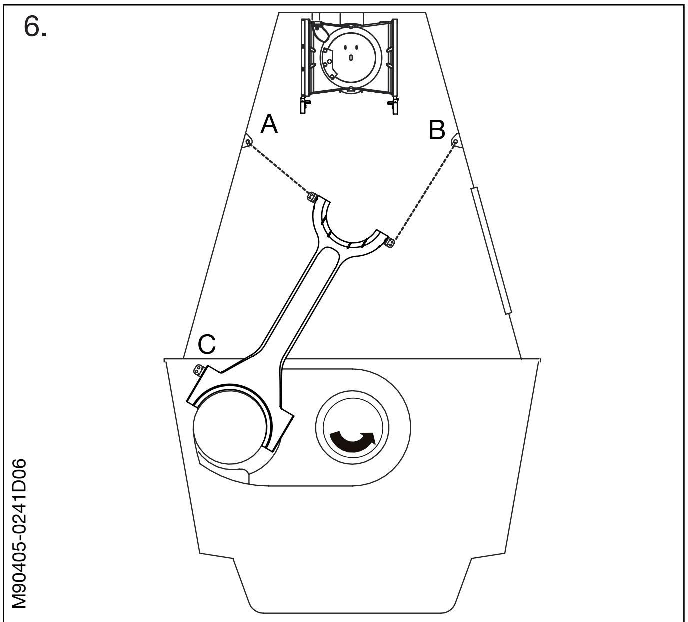
4.



5.



6.



4. Mount the four supports for guide shoes on the crosshead guides.

   Carefully turn the crank down towards the camshaft side, until the guide shoes rest on the supports.

   Adjust the support brackets to the guide shoes so that the weight of the crosshead is evenly distributed on the four supports.

5. Mount the lifting attachments for securing the connecting rod on the head of the connecting rod.

   Fasten tackles to lifting brackets **A** and **B** on the frame box wall, and attach the tackle hooks to the mentioned lifting attachments on the connecting rod head.

   Haul the tackles tight.

   Also mount a lifting attachment on the lower end of the connecting rod, on the exhaust side.

6. Turn the crankthrow carefully towards BDC while ‘following’ with the tackles, thus continuously supporting the connecting rod.

   The crosshead now rests on the four supports.

   Turn the crankthrow to 90° before BDC.

7. Shift tackle **B** from the lifting attachment on one side of the connecting rod to the lifting attachment on the other side.

   Dismount the lifting attachment on the camshaft side of the connecting rod.

   Attach a tackle to lifting bracket **A** on the frame box wall and connect the tackle hook to the lifting attachment on the lower end of the connecting rod.

   Mount the wire guide on the door frame.

   Turn the crankthrow towards TDC while 'following' with the tackles, thus continuously supporting and guiding the connecting rod towards the doorway.

8. Attach a tackle to the gallery-mounted lifting bracket **E**, and hook on to the lifting attachment on the connecting rod.

   Unhook tackle **A** from the lifting attachment on the head of the connecting rod.

   Turn the crank **carefully** upwards while 'following' with tackles **A**, **B** and **E**, guiding the head of the connecting rod out of the doorway.

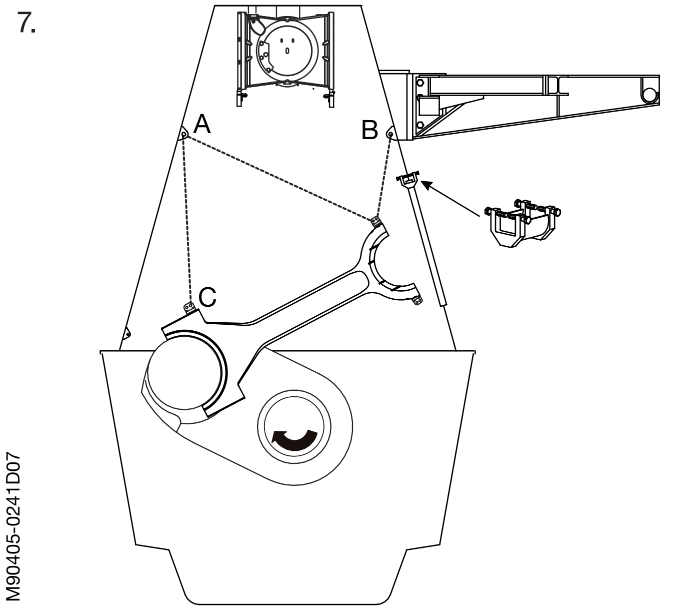
   Shift the tackles from one lifting attachment to the other as necessary.

9. Mount a strap around the connecting rod and suspend the connecting rod from the engine room crane. Shift tackle **B** from the lifting attachment on the head of the connecting rod to the lifting attachment at the bottom of the connecting rod.

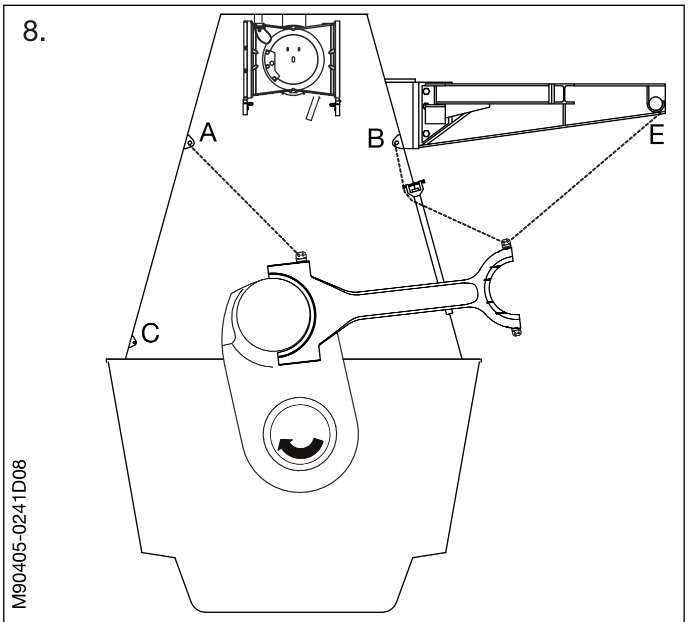
   Remove tackles **A** and **E**.

   Continue turning upwards till about 30° after TDC while 'following' with the tackles and the engine room crane.

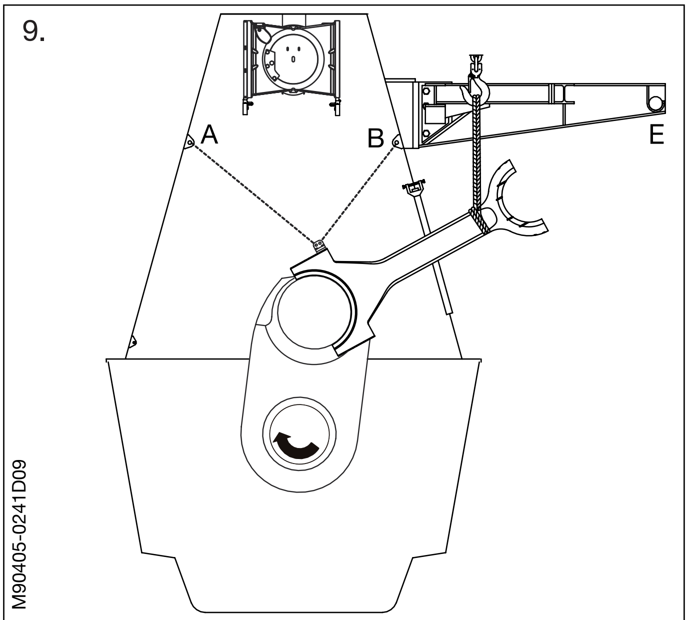
7.

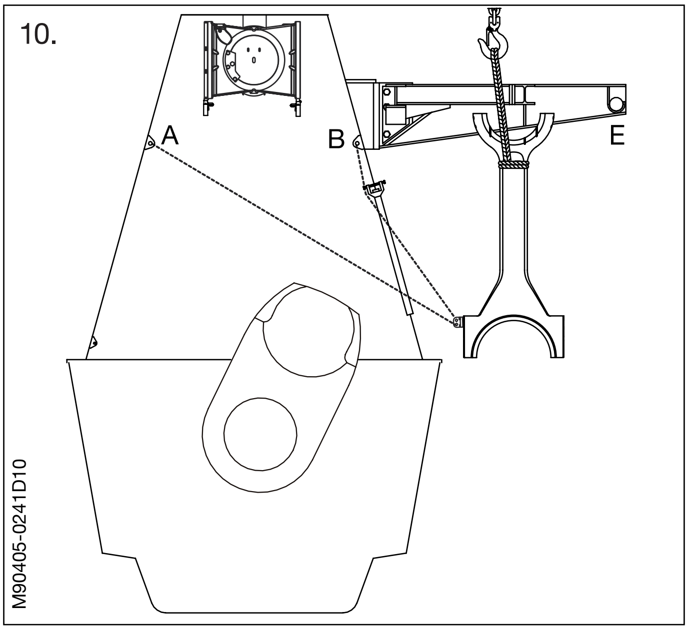


8.



9.





10. Lift the connecting rod out of the engine, using the tackles and the engine room crane.

    Remove the tackles, and lift the connecting rod away by means of the engine room crane.

1. Equip the connecting rod with the same lifting attachments as mentioned under dismantling.

   Turn the crank to a position about 25° past TDC on the camshaft side.

   Apply clean lubricating oil to the crankpin bearing shell and journal.

2. Carefully lift the connecting rod into the crankcase by alternate use of the engine room crane and the tackles attached to lifting brackets **A** and **B**.

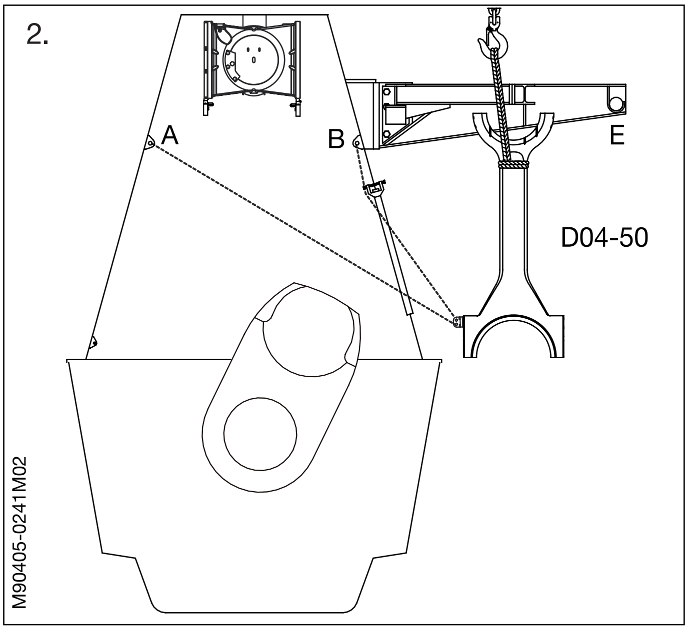
3. When the end of the connecting rod rests on the crankpin journal, follow with tackle **A**.

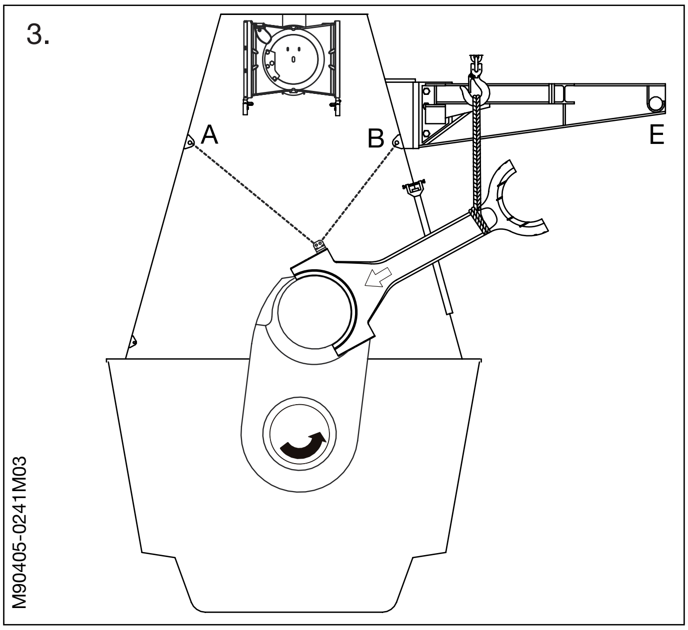
4. Turn the crankthrow towards BDC, past TDC, while ‘following’ with the tackles and the engine room crane.

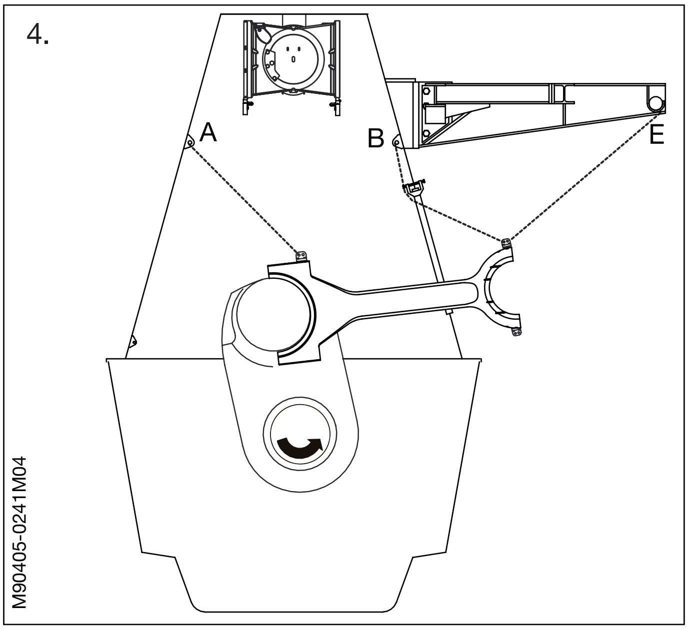
   Attach tackle **B** to the upper end of the connecting rod.

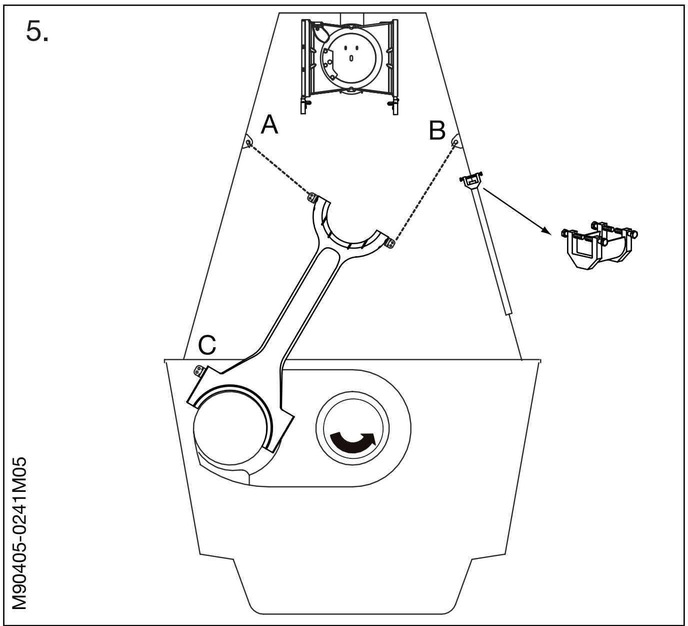
   Attach a tackle to bracket **E** and the upper end of the connecting rod.

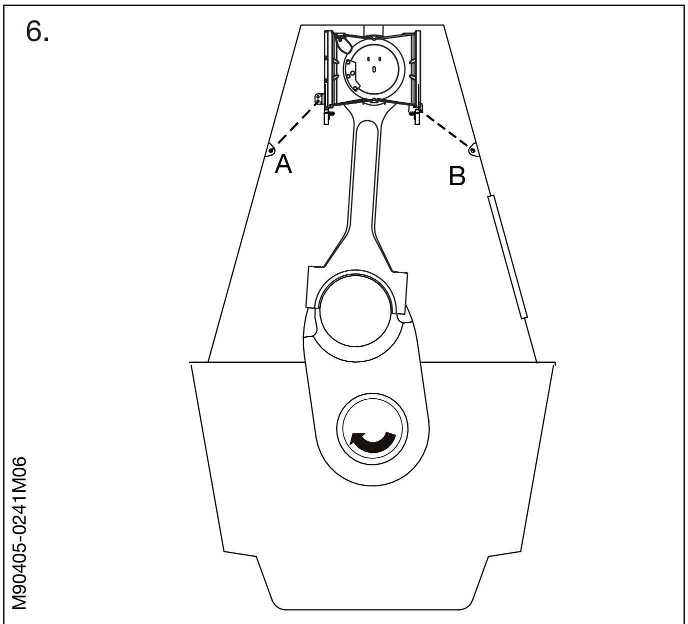
   Haul tight and remove the strap around the connecting rod.

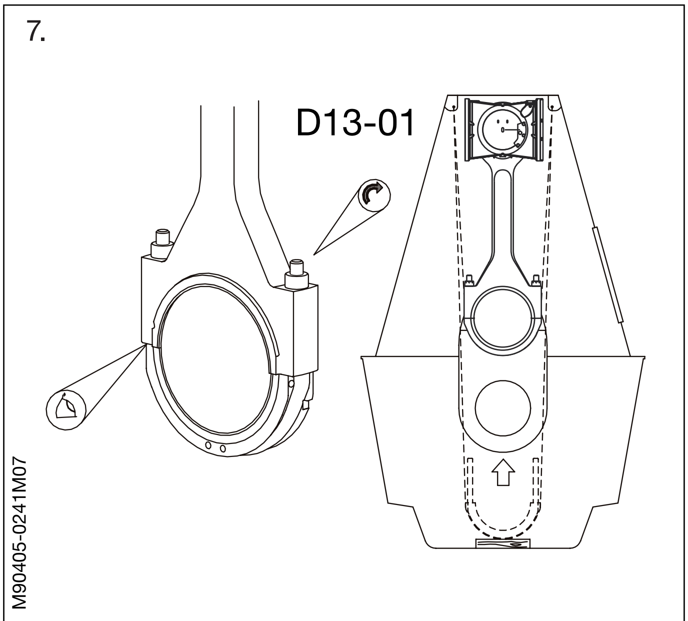












5. Turn the crankthrow to 90° before BDC.

   Shift tackle **A** from the lower end to the top of the connecting rod.

   Remove the tackle at **E**.

   Shift tackle **B** from the lifting attachment on one side to the other side of the connecting rod.

   Remove the wire guide on the door frame.

   Remove the lifting attachment at the lower end of the connecting rod.

6. Turn the crankthrow towards TDC while 'following' with the tackles.

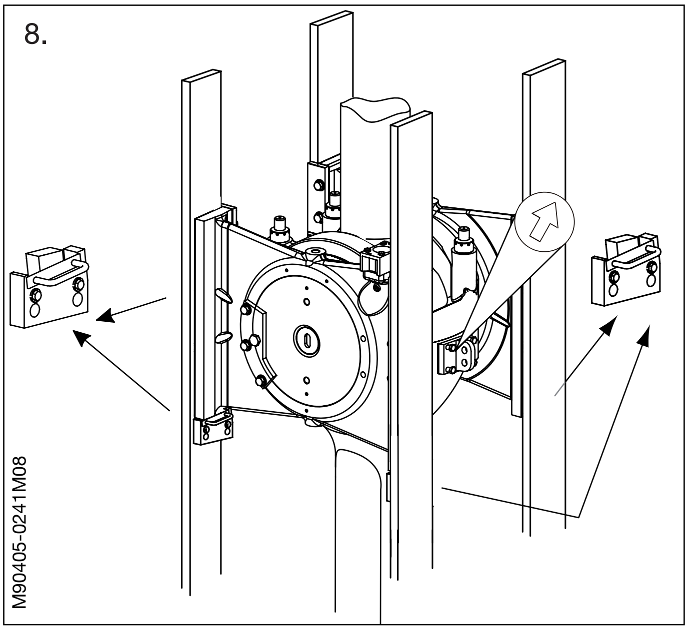
Caution!
Take care that the studs do not damage the crosshead bearing shell.

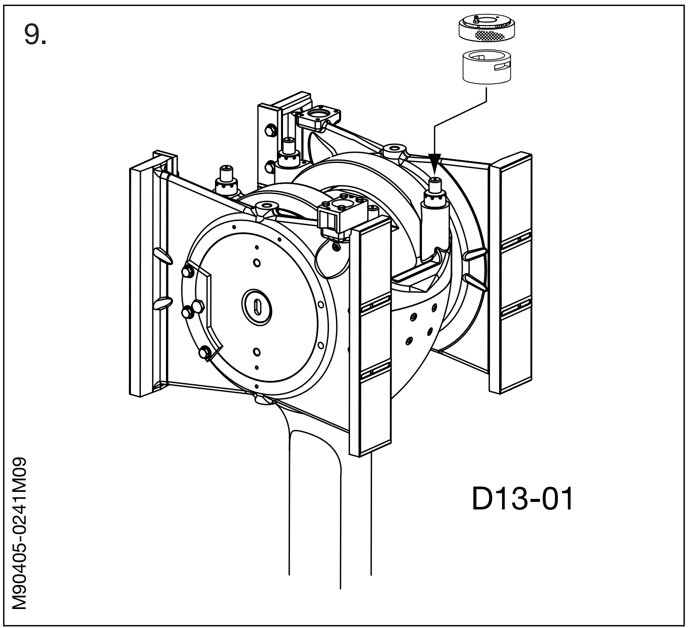
7. When the crank is in TDC, mount the crankpin bearing cap.
   See Procedure 904-4.4.

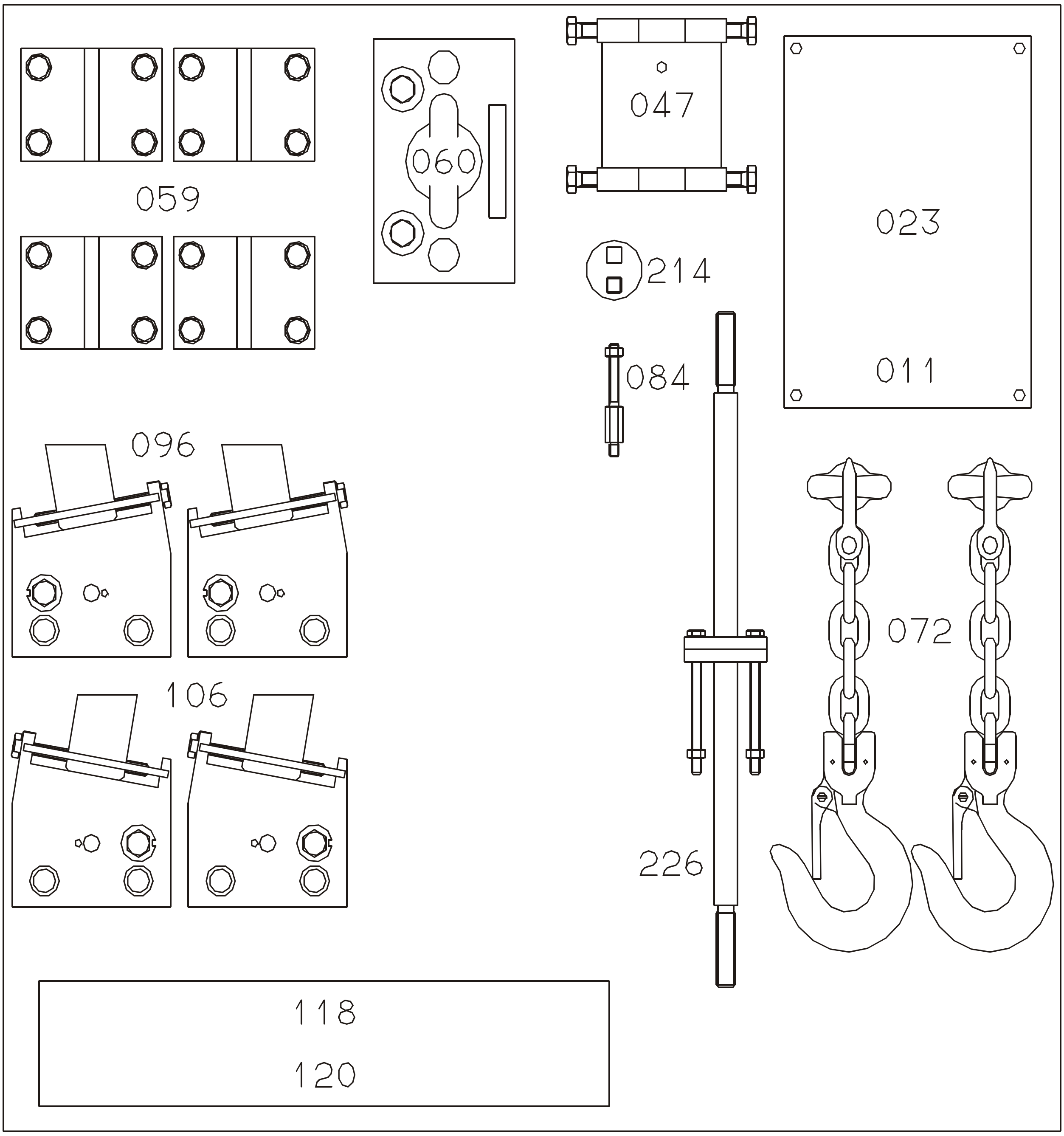
8. Remove the four supports from the cross-head guides and the lifting attachments from the connecting rod.

   Turn the crosshead down far enough to facilitate the tightening of the nuts.

9. Tighten all four crosshead bearing cap nuts simultaneously. *See Data.*





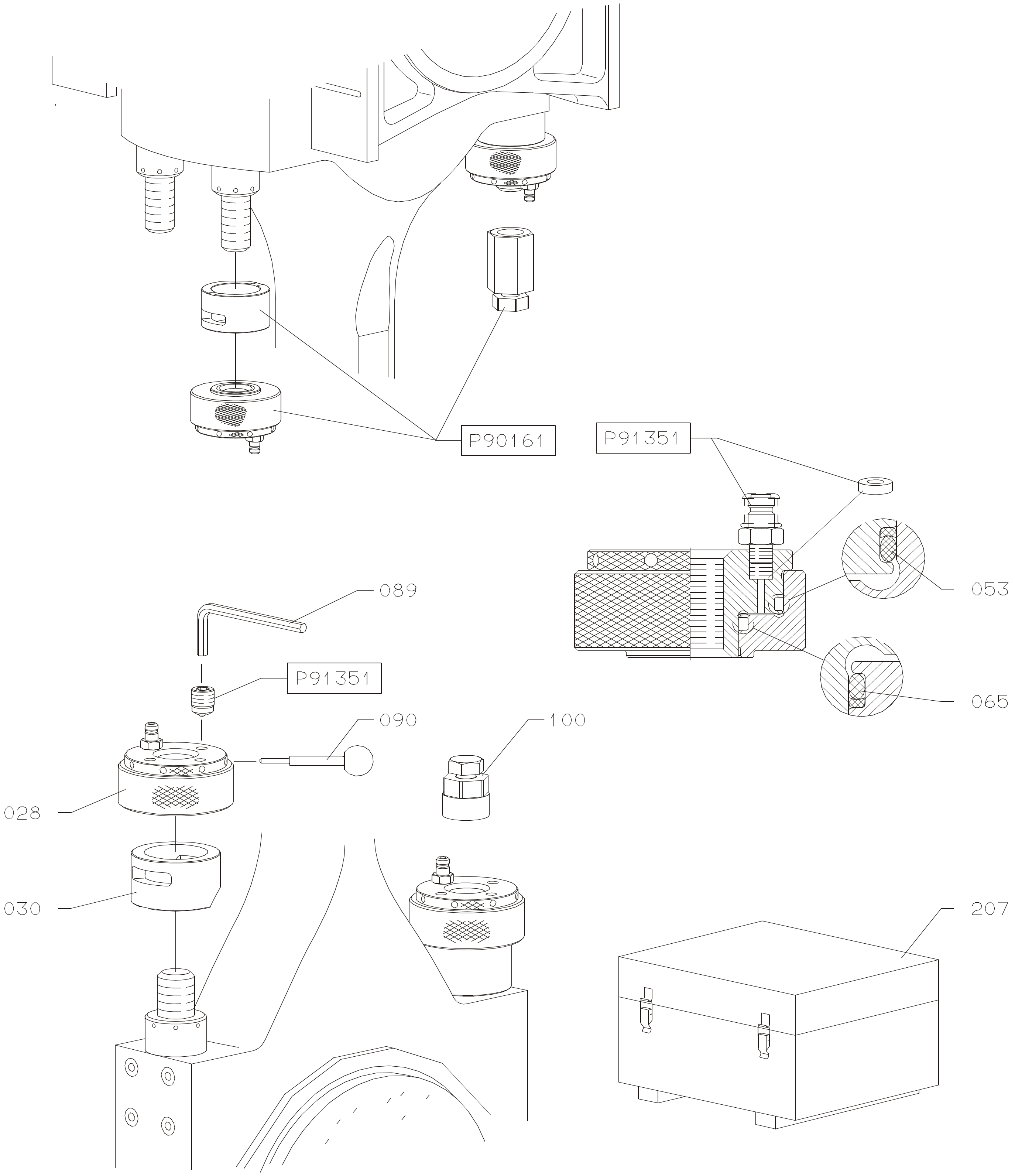

059
060
047
023
011
214
084
096
106
072
226
118
120

Connecting Rod and Crosshead Panel

Item No.	Item Description
011	Panel for tools
023	Name plate
047	Wire guide
059	Lifting attachment
060	Lifting tool for crosshead
072	Chain for suspending of piston
084	Retaining tool for telescope pipe
096	Support crosshead
106	Support crosshead
118	Rubber cover for crosshead
120	Rubber cover w. hole for crosshead
214	Torque wrench offset tool
251	Alignment tool

Item No.	Item Description

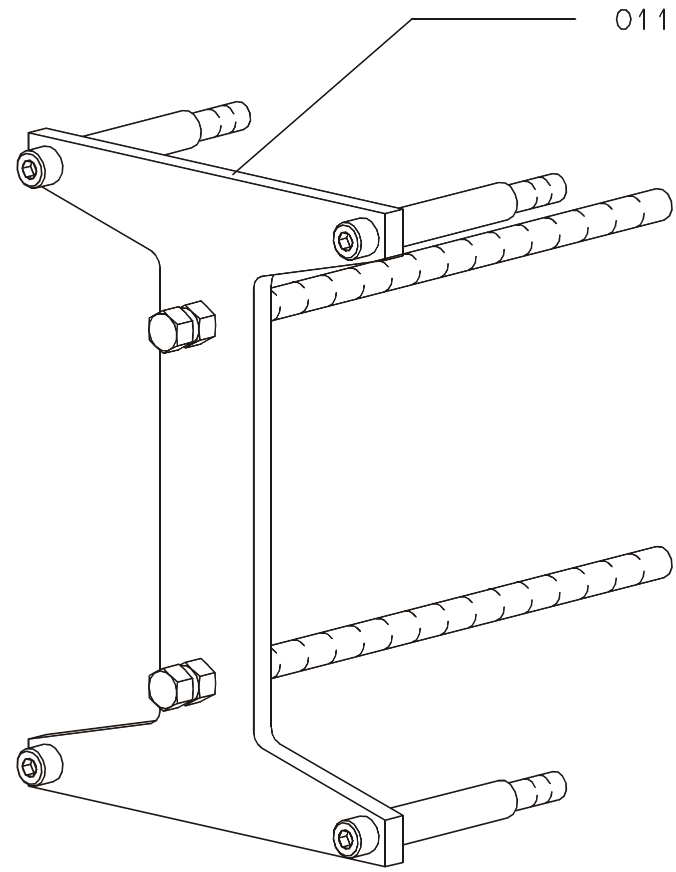
Notes:


P90161
P91351
089
P91351
090
100
028
030
053
065
207

Connecting Rod - Hydraulic Tools

Item No.	Item Description
028	Jack-hydraulic, complete
030	Support
053	Sealing ring with back-up ring
065	Sealing ring with back-up ring
089	Key, hexagon socket screw
090	Tommy bar
100	Stud setter
207	Hydraulic toolset, complete

Item No.	Item Description


011

Connecting Rod - Tools

Item No.	Item Description	Item No.	Item Description
011	Guide shoe extractor, complete		